

```

import pandas as pd
import matplotlib.pyplot as plt
import os

# Define the path to your "normal files" folder
folder_path = "normal files"

# Get a list of all CSV files in that directory
csv_files = [f for f in os.listdir(folder_path) if f.endswith('.csv')]

for file_name in csv_files:
    file_path = os.path.join(folder_path, file_name)
    print(f"Processing: {file_name}")

    try:
        # 1. Read metadata (first 25 rows)
        header_df = pd.read_csv(file_path, nrows=25, header=None, usecols=[0, 1])
        metadata = dict(zip(header_df[0], header_df[1]))

        # 2. Read waveform data (skip header)
        df = pd.read_csv(file_path, skiprows=26, header=None, usecols=[0, 1])
        df.columns = ['Time', 'Voltage']

        # 3. Create the Plot
        plt.figure(figsize=(10, 5))
        plt.plot(df['Time'], df['Voltage'], color='tab:green', linewidth=1)

        # Metadata extraction
        source = metadata.get('Source', 'Unknown')
        v_scale = metadata.get('Vertical Scale', 'N/A')

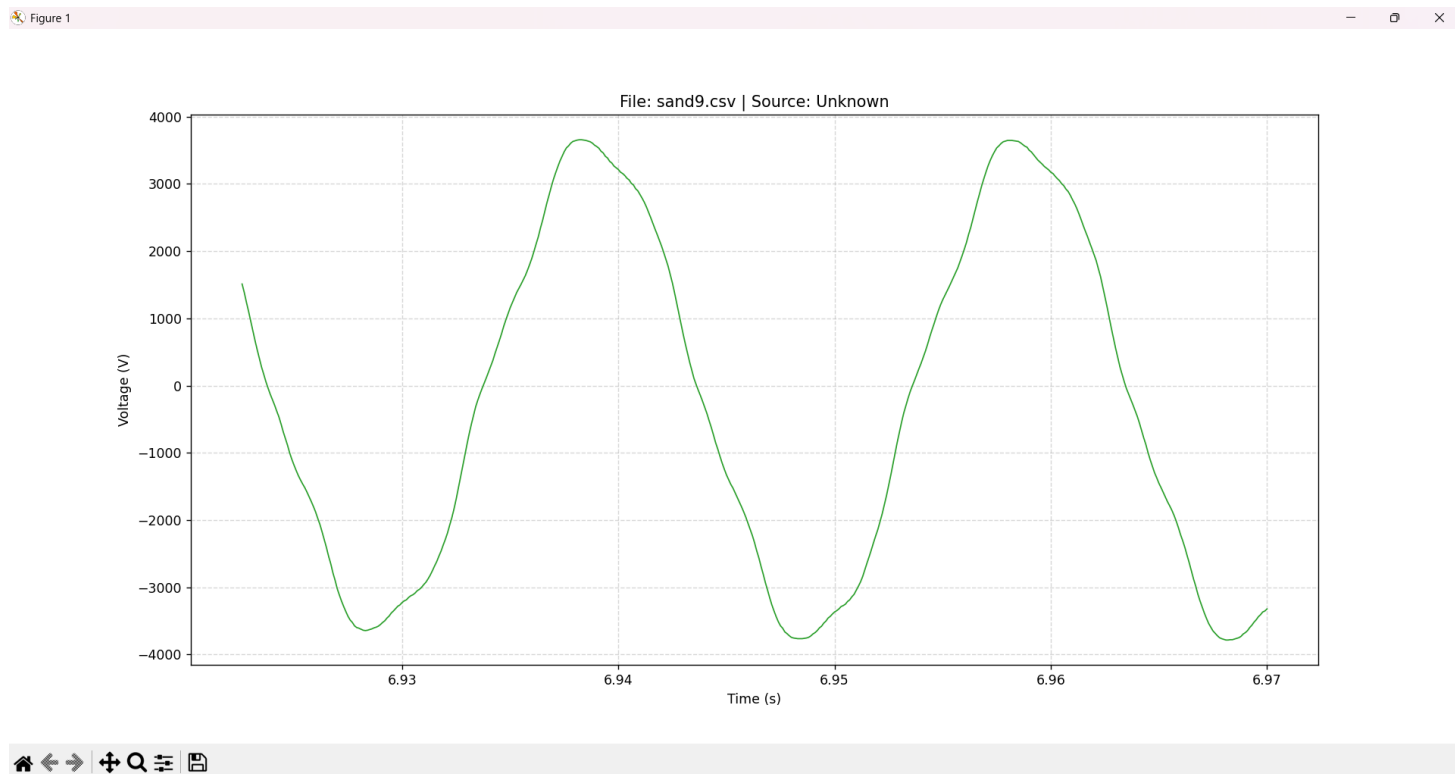
        plt.title(f"File: {file_name} | Source: {source}", fontsize=12)
        plt.xlabel("Time (s)")
        plt.ylabel("Voltage (V)")
        plt.grid(True, linestyle='--', alpha=0.5)

        # Optional: Save the plots instead of just showing them
        # plt.savefig(f"{file_name}.png")

        plt.show()

    except Exception as e:
        print(f"Could not process {file_name}: {e}")

```



```

import pandas as pd
import matplotlib.pyplot as plt
import numpy as np
import os

# --- CONFIGURATION ---
folder_path = "normal files"
target_freqs = [50, 100, 150, 200, 250, 300, 350, 400, 450, 500]

```

```

# Get all CSV files from the folder
csv_files = [f for f in os.listdir(folder_path) if f.endswith('.csv')]

def get_genuine_fft(dataframe):
    # Calculate sampling period (dt) directly from the data
    # This is the most critical step for accuracy
    t = dataframe['Time'].values
    dt = np.mean(np.diff(t))

    # Genuine signal: Remove DC offset
    sig = dataframe['Voltage'].values - np.mean(dataframe['Voltage'].values)
    n = len(sig)

    # Raw FFT (No windowing, No padding)
    fft_vals = np.fft.rfft(sig)
    fft_mag = np.abs(fft_vals) * (2.0 / n)
    freqs = np.fft.rfftfreq(n, d=dt)

    return freqs, fft_mag, dt

def print_coordinates(file_name, freqs, magnitudes):
    total_mag = np.sum(magnitudes)
    print(f"\n--- GENUINE COORDINATES: {file_name} ---")
    print(f"{'Target (Hz)':<12} | {'Actual (Hz)':<12} | {'Mag (V)':<12} | {'Contrib (%)':<12}")
    print("-" * 65)
    for t in target_freqs:
        idx = np.abs(freqs - t).argmin()
        actual_f = freqs[idx]
        mag = magnitudes[idx]
        contribution = (mag / total_mag) * 100
        print(f"{'t':<12} | {'actual_f':<12.2f} | {'mag':<12.4f} | {'contribution':<12.2f}%")

# --- EXECUTION LOOP ---
for file_name in csv_files:
    file_path = os.path.join(folder_path, file_name)

    try:
        # 1. Load Data (Adjust skiprows if these files don't have the 26-line header)
        # If these are 'normal' CSVs without headers, use skiprows=0
        df = pd.read_csv(file_path, skiprows=26, header=None, usecols=[0, 1])
        df.columns = ['Time', 'Voltage']

        # 2. Process
        f_all, m_all, actual_dt = get_genuine_fft(df)
        print_coordinates(file_name, f_all, m_all)
        print(f"Detected Sampling Rate: {1 / actual_dt:.2f} Hz")

        # 3. Plotting
        fig, (ax1, ax2) = plt.subplots(2, 1, figsize=(12, 10), layout="constrained")

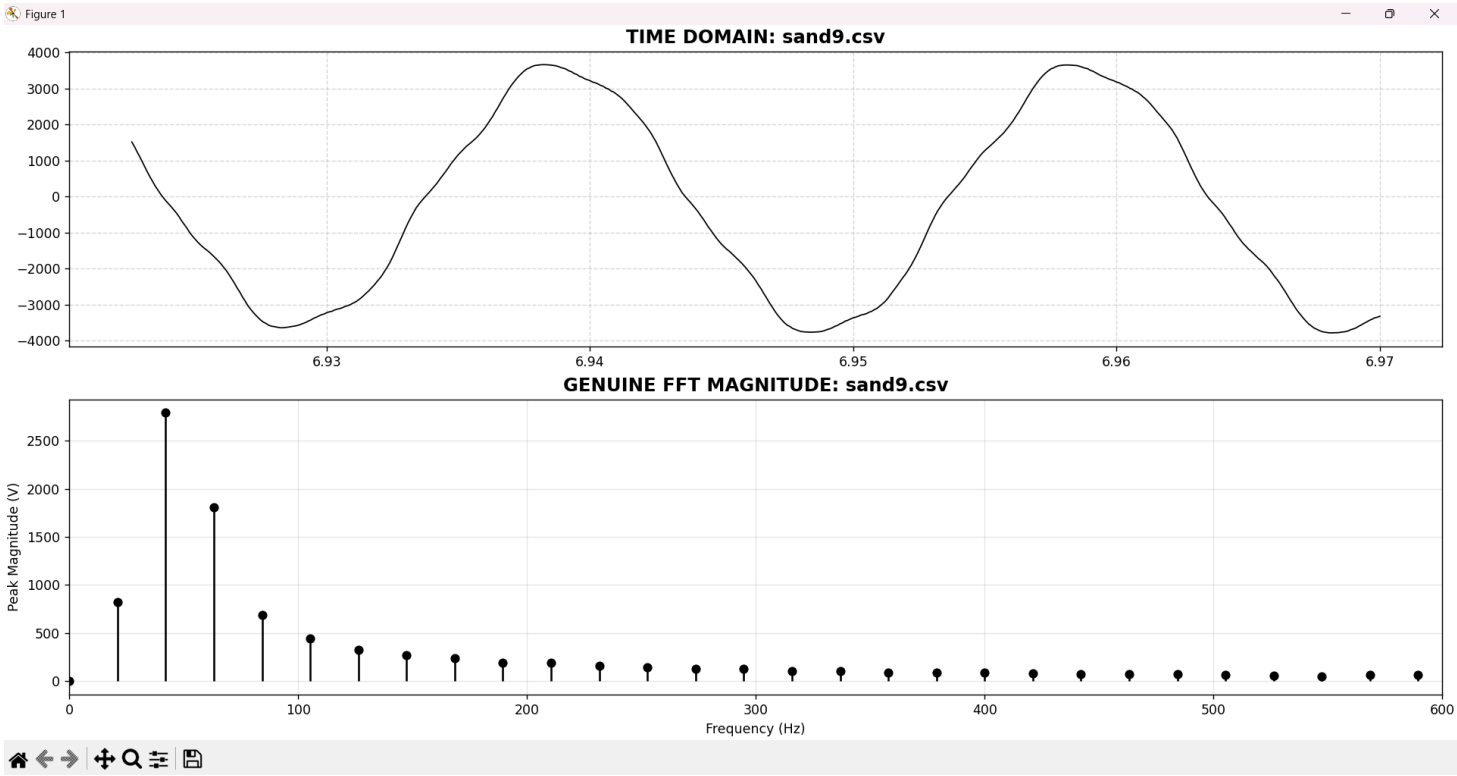
        # Time Domain
        ax1.plot(df['Time'], df['Voltage'], color='black', linewidth=1)
        ax1.set_title(f"TIME DOMAIN: {file_name}", fontsize=14, fontweight='bold')
        ax1.grid(True, linestyle='--', alpha=0.5)

        # Discrete FFT
        ax2.stem(f_all, m_all, linefmt='black', markerfmt='ko', basefmt=' ')
        ax2.set_xlim(0, 600)
        ax2.set_title(f"GENUINE FFT MAGNITUDE: {file_name}", fontsize=14, fontweight='bold')
        ax2.set_xlabel("Frequency (Hz)")
        ax2.set_ylabel("Peak Magnitude (V)")
        ax2.grid(True, alpha=0.3)

        plt.show()

    except Exception as e:
        print(f"Skipping {file_name} due to error: {e}")

```

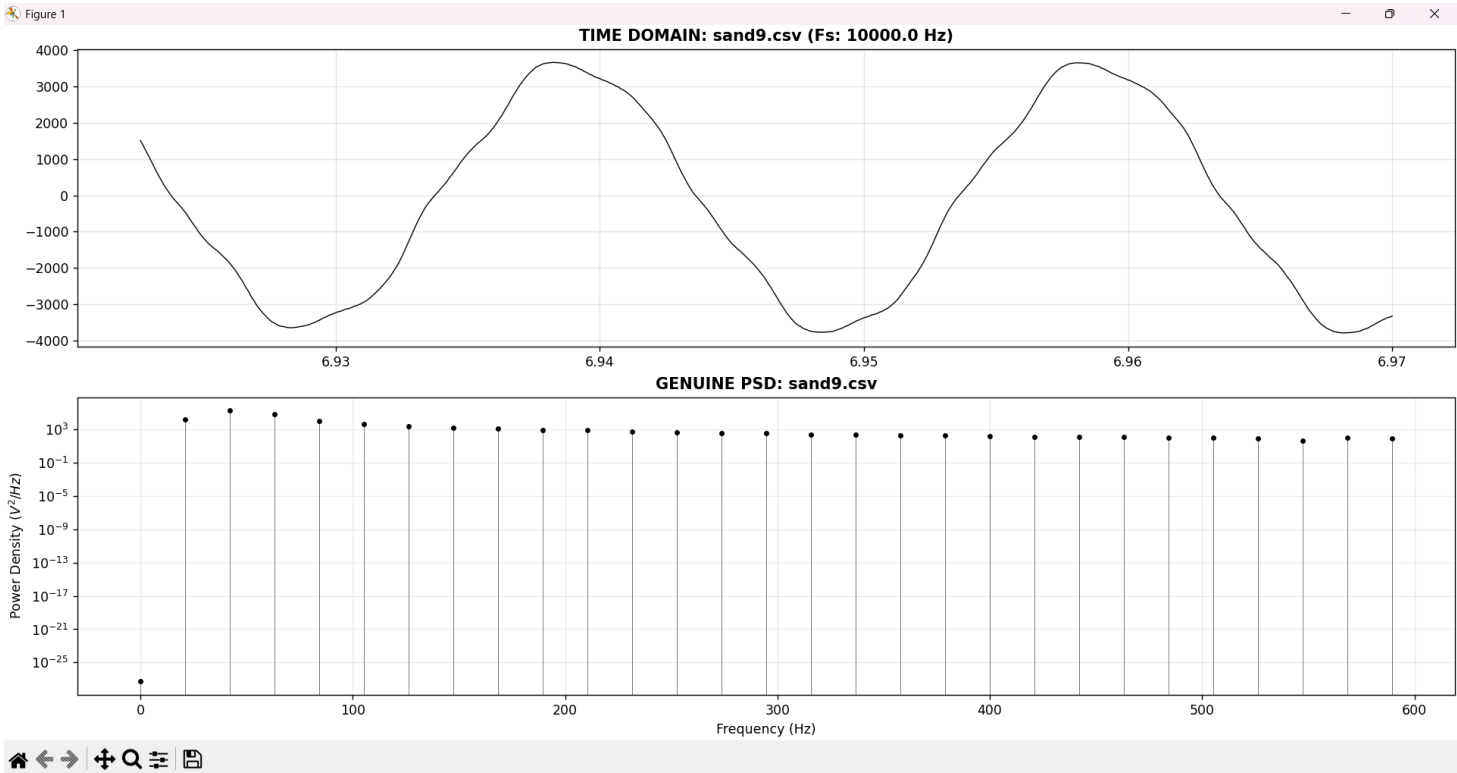


--- GENUINE COORDINATES: sand9.csv ---

Target (Hz) | Actual (Hz) | Mag (V) | Contrib (%)

50	42.11	2790.7455	21.38	%
100	105.26	438.2989	3.36	%
150	147.37	266.9769	2.04	%
200	189.47	190.2187	1.46	%
250	252.63	143.3725	1.10	%
300	294.74	123.9488	0.95	%
350	357.89	90.0326	0.69	%
400	400.00	84.2312	0.65	%
450	442.11	74.0674	0.57	%
500	505.26	66.8016	0.51	%

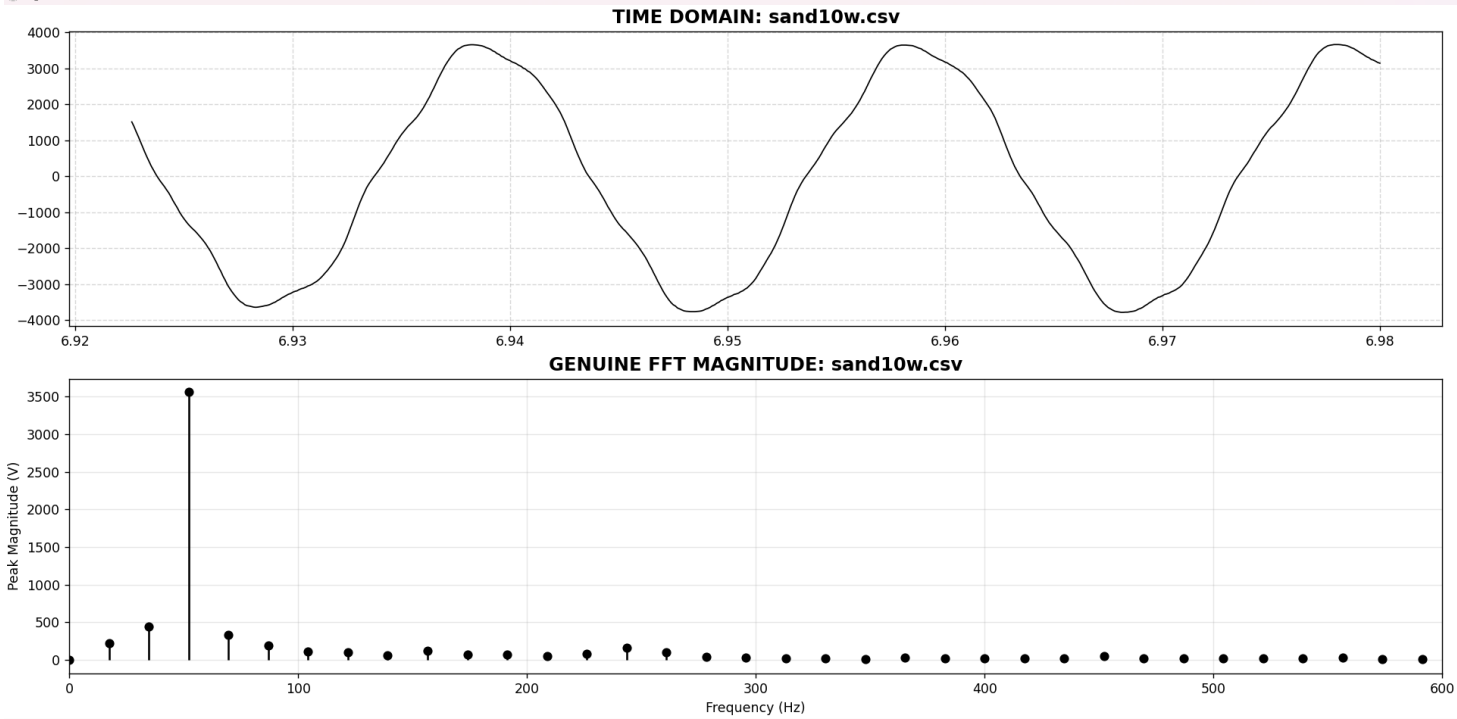
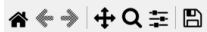
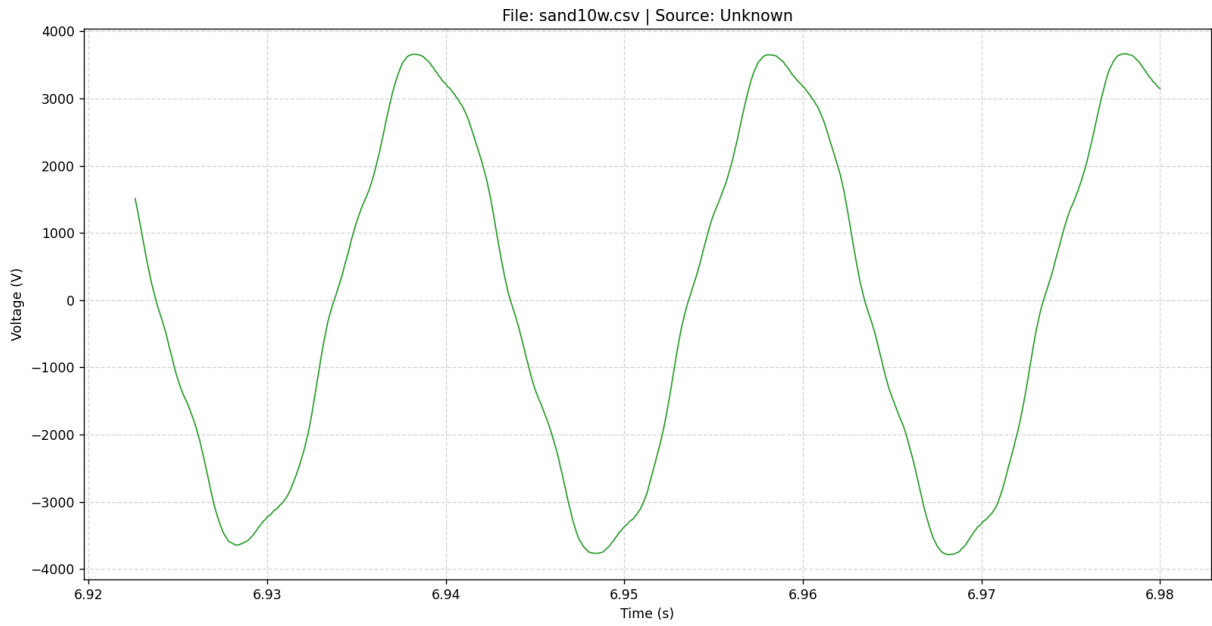
Detected Sampling Rate: 10000.00 Hz



--- EXACT POWER CONTRIBUTION: sand9.csv ---

Freq (Hz) | Power (V^2/Hz) | Contribution (%)

42.1	1.85e+05	60.52	%
105.3	4.56e+03	1.49	%
147.4	1.69e+03	0.55	%
189.5	8.59e+02	0.28	%
252.6	4.88e+02	0.16	%
294.7	3.65e+02	0.12	%
357.9	1.93e+02	0.06	%
400.0	1.69e+02	0.06	%
442.1	1.30e+02	0.04	%
505.3	1.06e+02	0.03	%

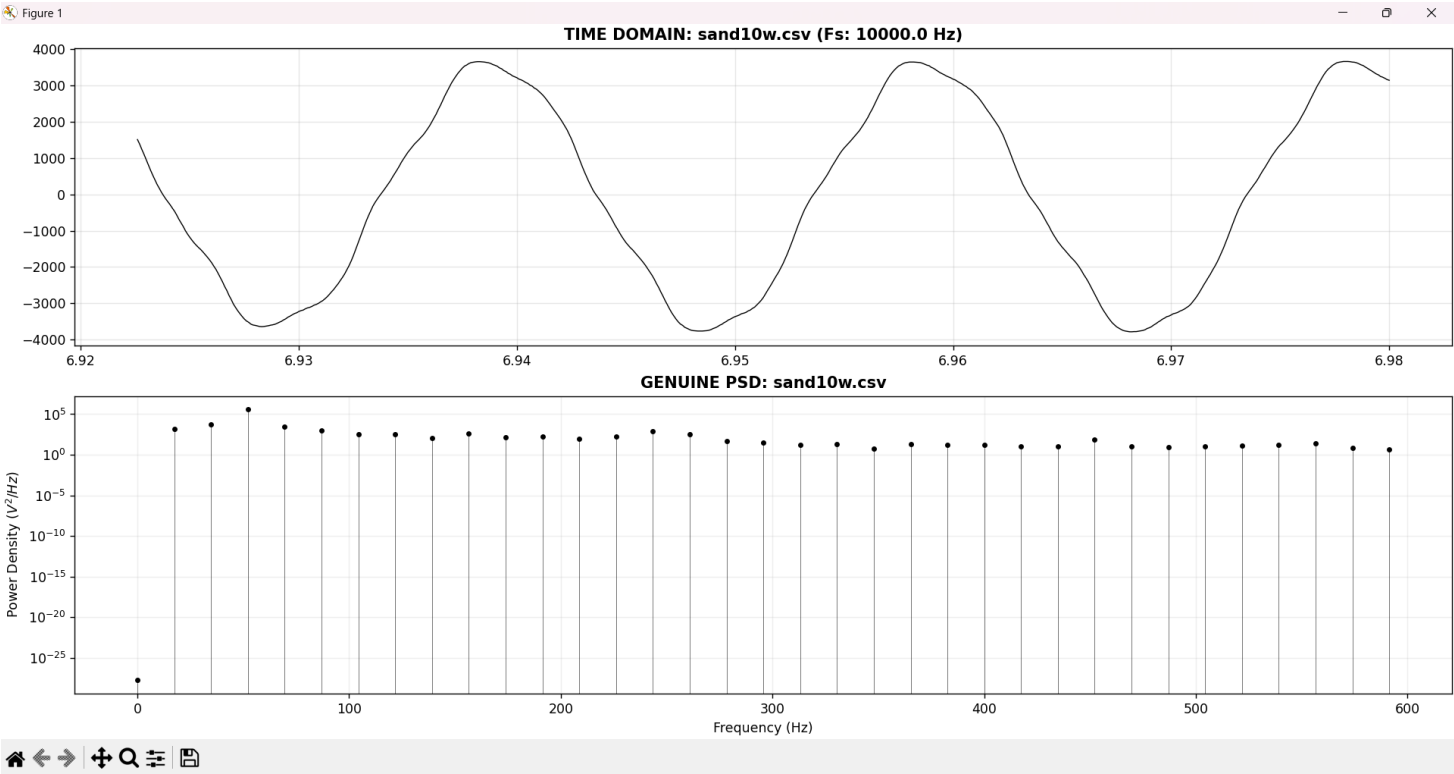


--- GENUINE COORDINATES: sand10w.csv ---

Target (Hz) | Actual (Hz) | Mag (V) | Contrib (%)

50	52.17	3561.6153	48.80	%
100	104.35	113.3695	1.55	%
150	156.52	117.9009	1.62	%
200	208.70	54.1857	0.74	%
250	243.48	157.0741	2.15	%
300	295.65	33.5854	0.46	%
350	347.83	14.0682	0.19	%
400	400.00	23.5280	0.32	%
450	452.17	51.3716	0.70	%
500	504.35	18.7378	0.26	%

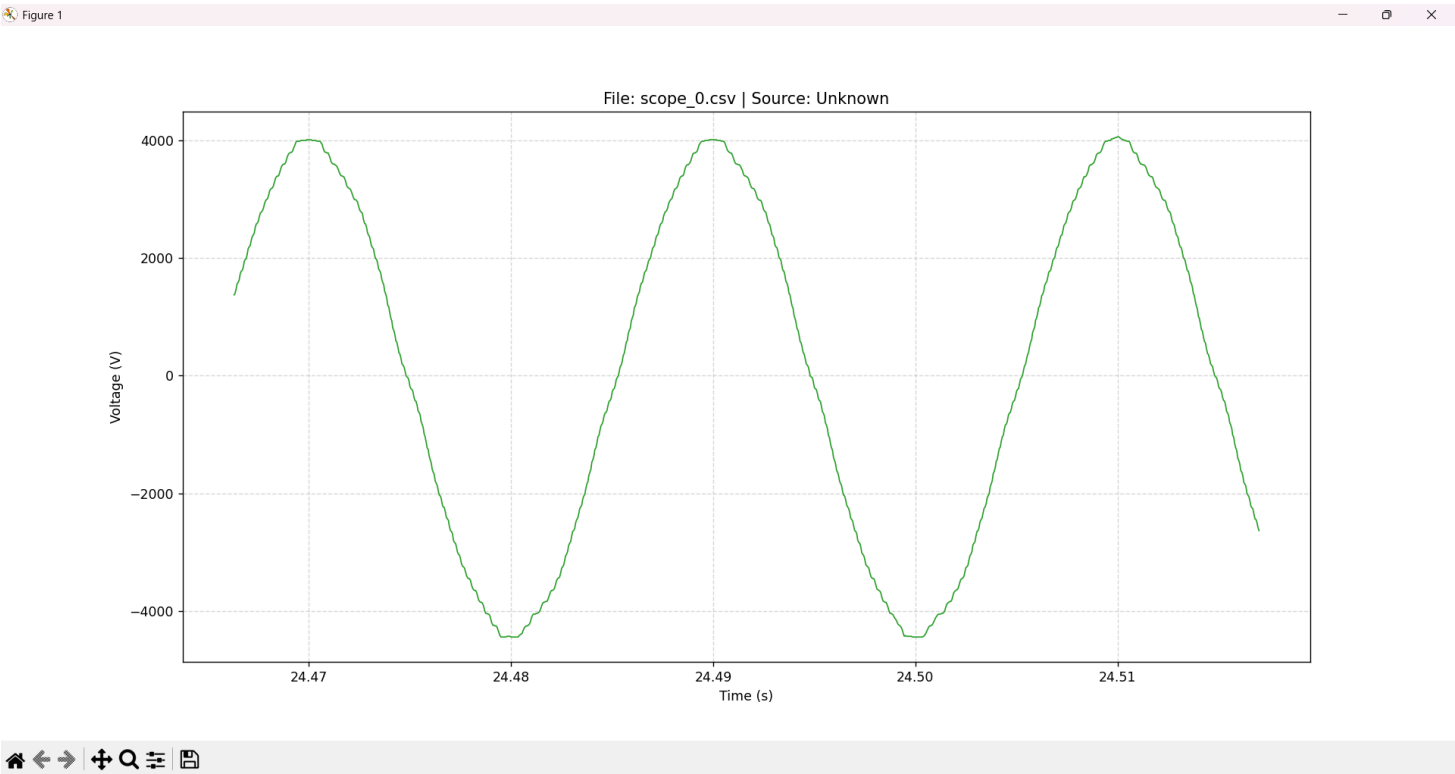
Detected Sampling Rate: 10000.00 Hz

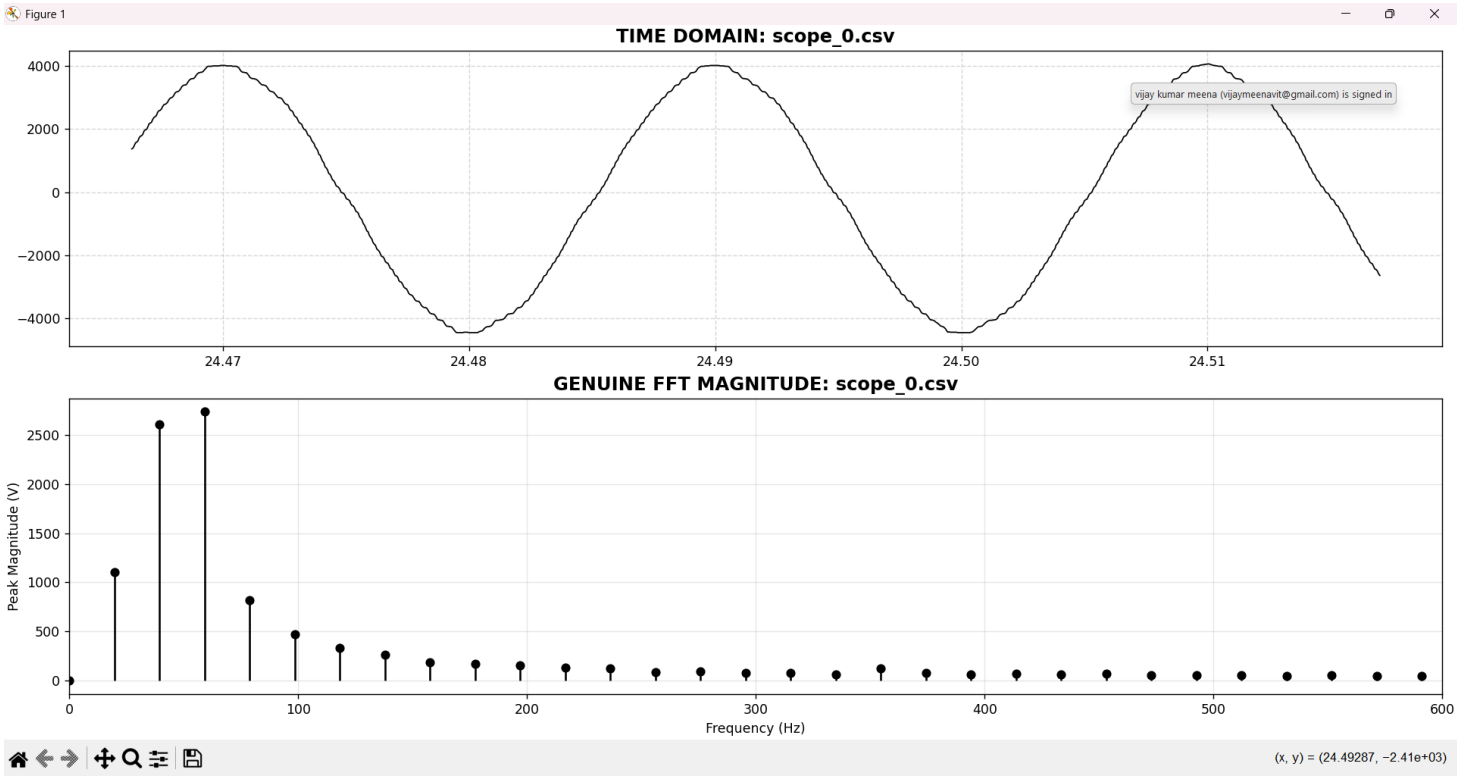


--- EXACT POWER CONTRIBUTION: sand10w.csv ---

Freq (Hz) | Power (V^2/Hz) | Contribution (%)

52.2	3.65e+05	96.22	%
104.3	3.70e+02	0.10	%
156.5	4.00e+02	0.11	%
191.3	1.61e+02	0.04	%
243.5	7.09e+02	0.19	%
295.7	3.24e+01	0.01	%
347.8	5.69e+00	0.00	%
400.0	1.59e+01	0.00	%
452.2	7.59e+01	0.02	%
504.3	1.01e+01	0.00	%



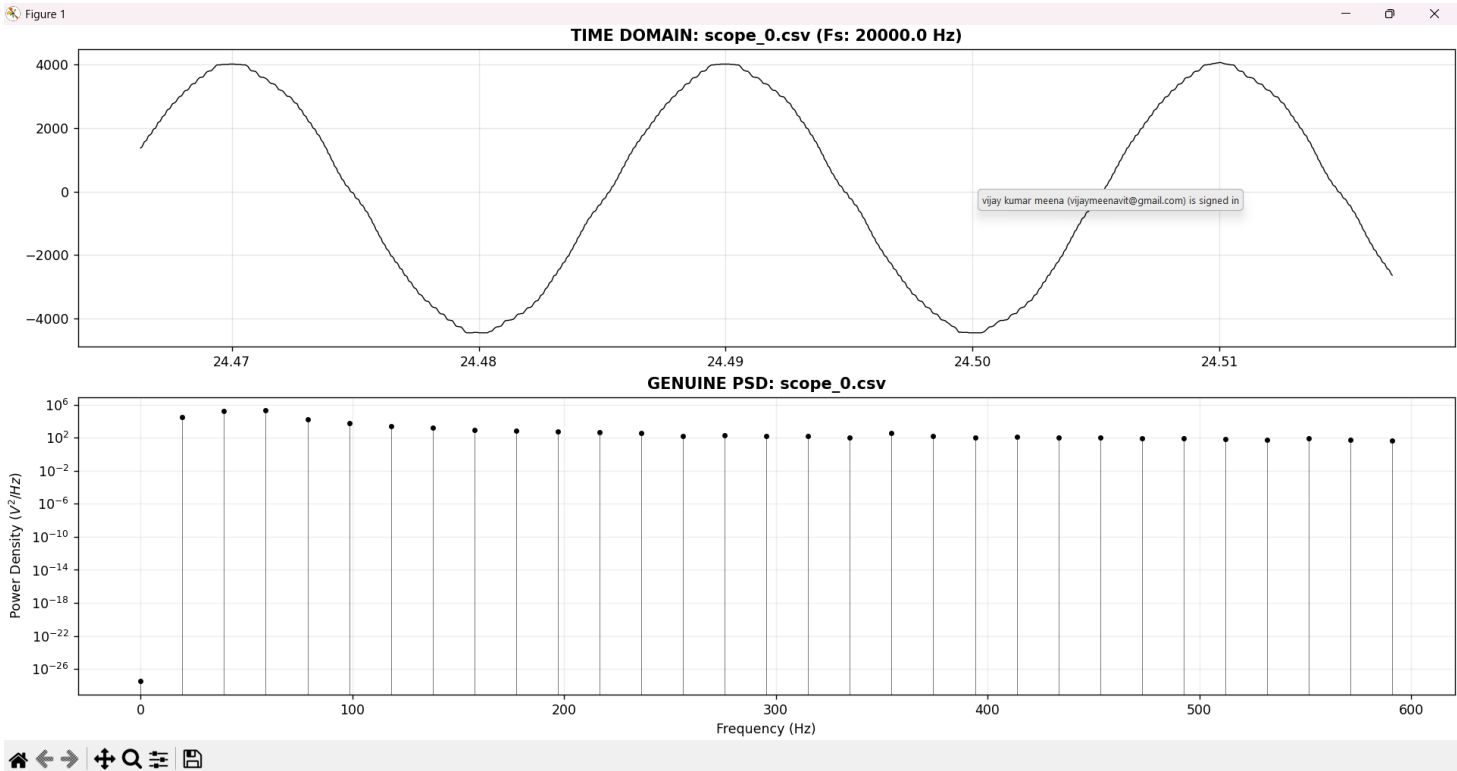


--- GENUINE COORDINATES: scope_0.csv ---

Target (Hz) | Actual (Hz) | Mag (V) | Contrib (%)

50	59.11	2738.0958	19.27	%
100	98.52	472.8630	3.33	%
150	157.64	182.2616	1.28	%
200	197.04	150.8510	1.06	%
250	256.16	83.1341	0.59	%
300	295.57	78.8146	0.55	%
350	354.68	124.2081	0.87	%
400	394.09	63.6822	0.45	%
450	453.20	65.9427	0.46	%
500	492.61	56.1506	0.40	%

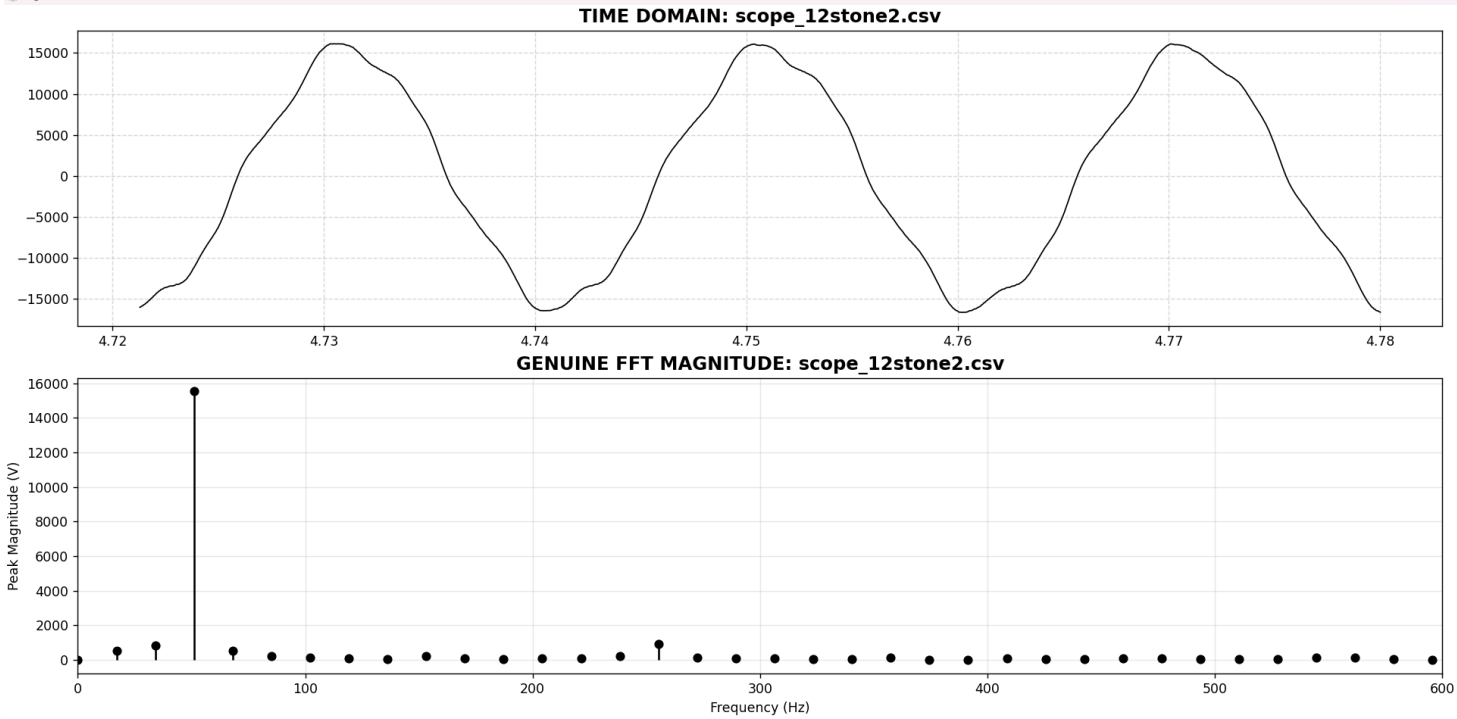
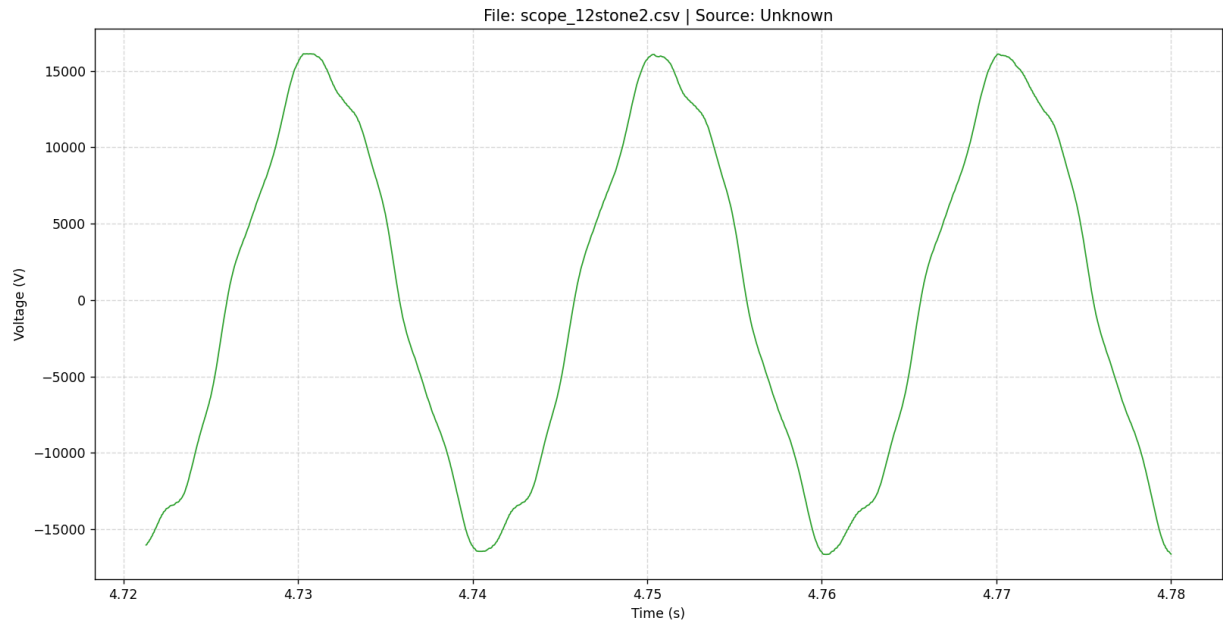
Detected Sampling Rate: 20000.00 Hz



--- EXACT POWER CONTRIBUTION: scope_0.csv ---

Freq (Hz) | Power (V^2/Hz) | Contribution (%)

59.1	1.90e+05	44.57	%
98.5	5.67e+03	1.33	%
157.6	8.43e+02	0.20	%
197.0	5.77e+02	0.14	%
256.2	1.75e+02	0.04	%
295.6	1.58e+02	0.04	%
354.7	3.91e+02	0.09	%
394.1	1.03e+02	0.02	%
453.2	1.10e+02	0.03	%
492.6	8.00e+01	0.02	%

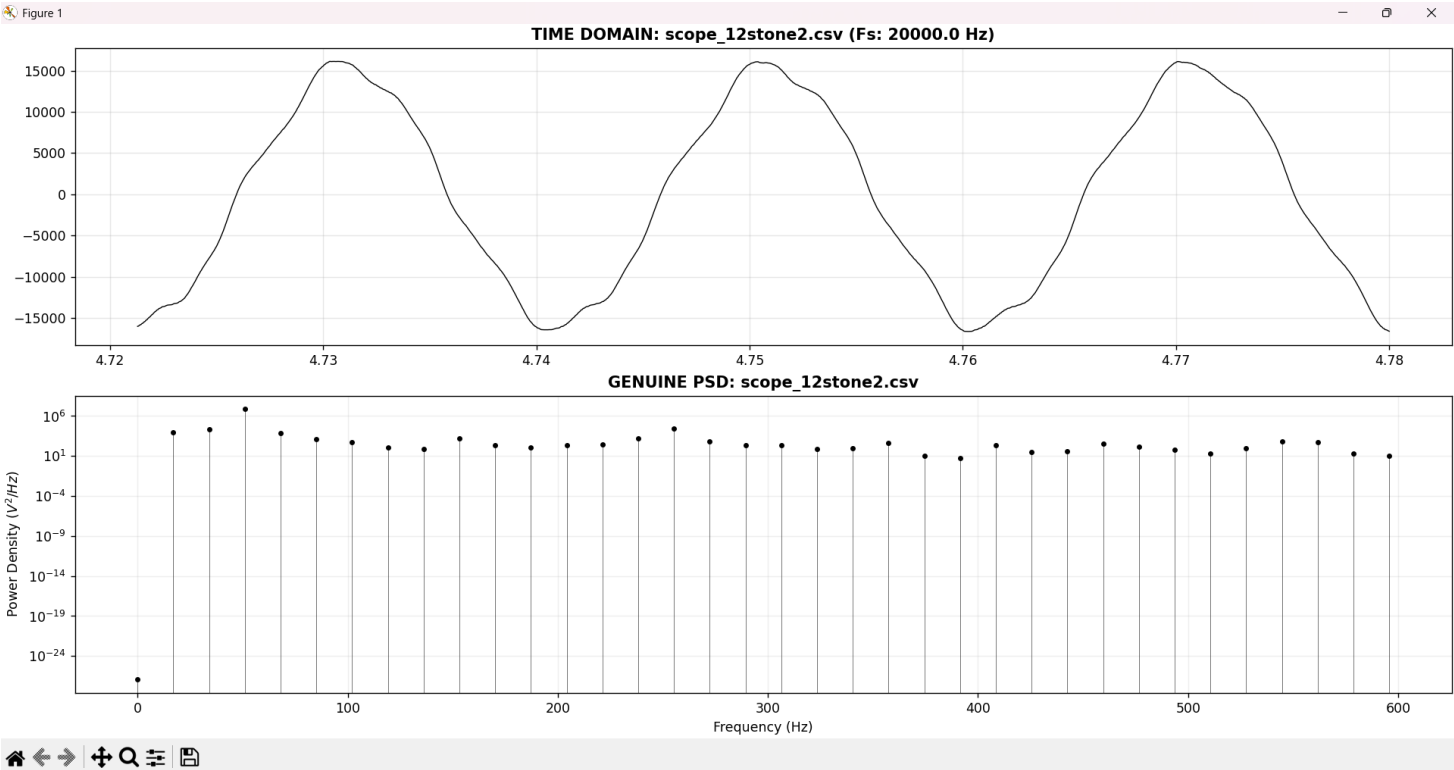


--- GENUINE COORDINATES: scope_12stone2.csv ---

Target (Hz) | Actual (Hz) | Mag (V) | Contrib (%)

50	51.06	15526.2648	70.81	%
100	102.13	141.2986	0.64	%
150	153.19	236.0406	1.08	%
200	204.26	84.0289	0.38	%
250	255.32	914.7194	4.17	%
300	306.38	91.3758	0.42	%
350	357.45	126.5781	0.58	%
400	391.49	13.7087	0.06	%
450	442.55	34.7423	0.16	%
500	493.62	43.1724	0.20	%

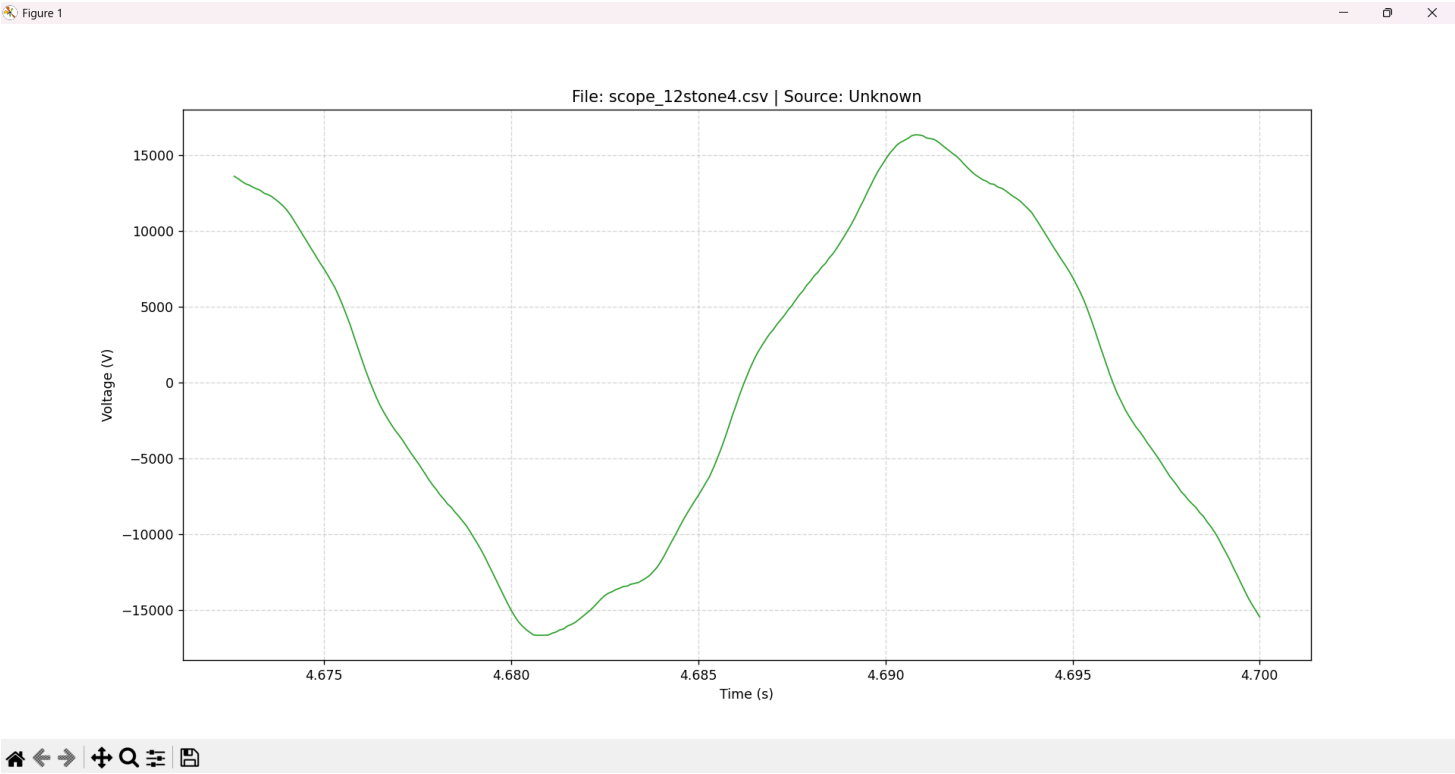
Detected Sampling Rate: 20000.00 Hz

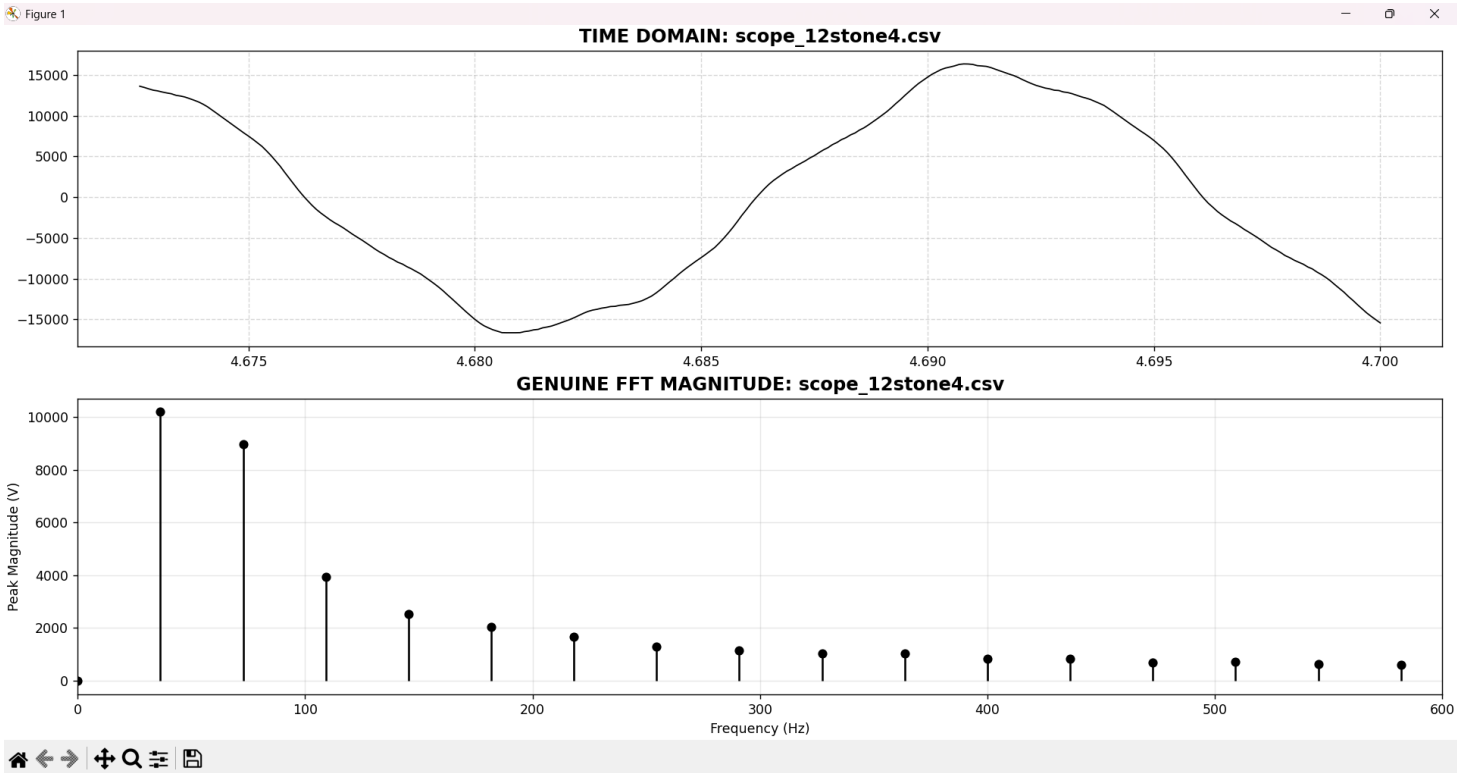


--- EXACT POWER CONTRIBUTION: scope_12stone2.csv ---

Freq (Hz) | Power (V^2/Hz) | Contribution (%)

51.1	7.08e+06	98.99	%
102.1	5.86e+02	0.01	%
153.2	1.64e+03	0.02	%
204.3	2.07e+02	0.00	%
255.3	2.46e+04	0.34	%
306.4	2.45e+02	0.00	%
357.4	4.71e+02	0.01	%
391.5	5.52e+00	0.00	%
442.6	3.55e+01	0.00	%
493.6	5.48e+01	0.00	%



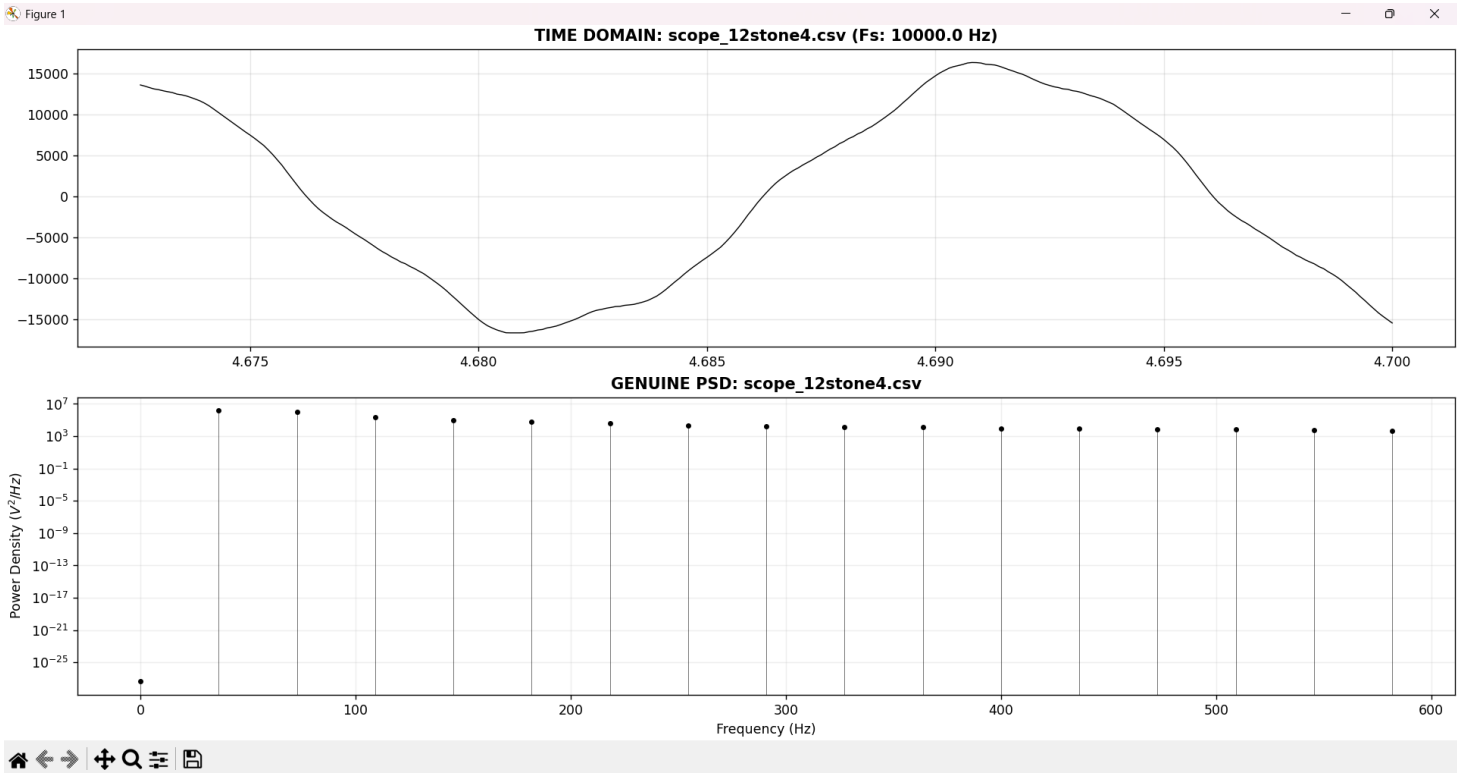


--- GENUINE COORDINATES: scope_12stone4.csv ---

Target (Hz) | Actual (Hz) | Mag (V) | Contrib (%)

50	36.36	10202.1710	16.94	%
100	109.09	3940.8892	6.54	%
150	145.45	2523.1550	4.19	%
200	218.18	1675.6983	2.78	%
250	254.55	1295.1990	2.15	%
300	290.91	1150.5857	1.91	%
350	363.64	1026.2922	1.70	%
400	400.00	835.1649	1.39	%
450	436.36	819.4424	1.36	%
500	509.09	715.2868	1.19	%

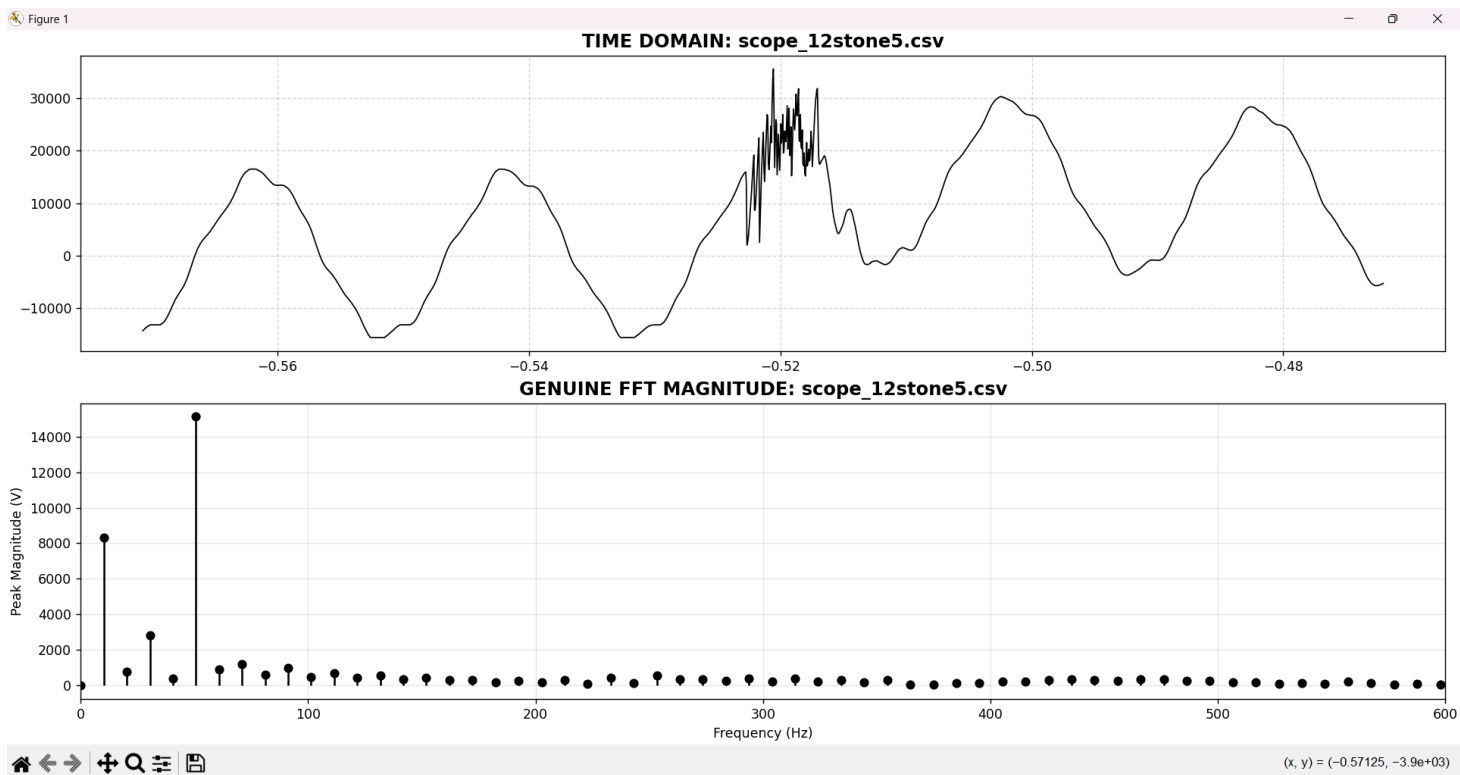
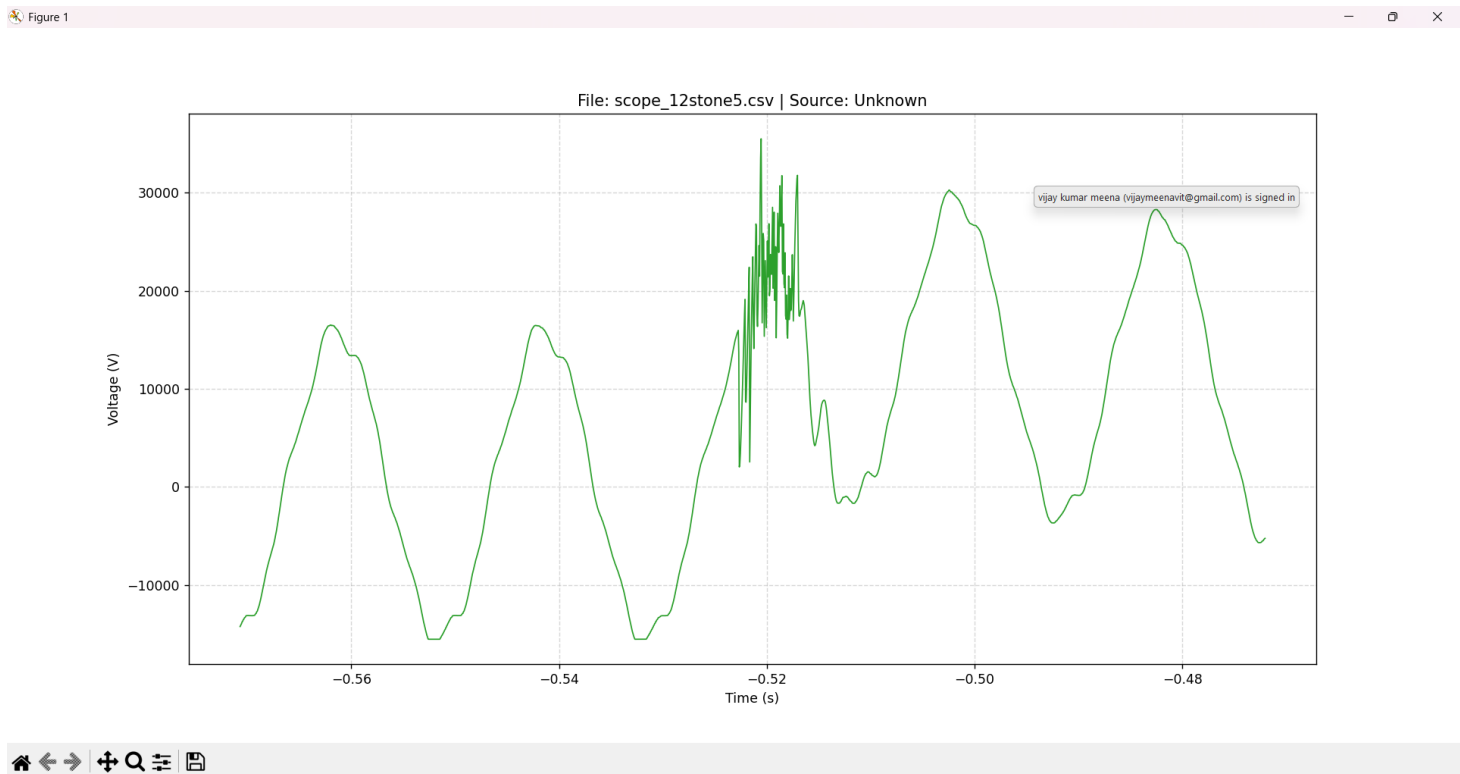
Detected Sampling Rate: 10000.00 Hz



--- EXACT POWER CONTRIBUTION: scope_12stone4.csv ---

Freq (Hz) | Power (V^2/Hz) | Contribution (%)

36.4	1.43e+06	46.92	%
109.1	2.14e+05	7.00	%
145.5	8.75e+04	2.87	%
181.8	5.80e+04	1.90	%
254.5	2.31e+04	0.76	%
290.9	1.82e+04	0.60	%
363.6	1.45e+04	0.47	%
400.0	9.59e+03	0.31	%
436.4	9.23e+03	0.30	%
509.1	7.03e+03	0.23	%



--- GENUINE COORDINATES: scope_12stone5.csv ---

Target (Hz) | Actual (Hz) | Mag (V) | Contrib (%)

50	50.66	15128.7821	18.08	%
100	101.32	452.6222	0.54	%
150	151.98	406.3727	0.49	%
200	202.63	163.6994	0.20	%
250	253.29	561.2805	0.67	%
300	303.95	214.9990	0.26	%
350	354.61	277.1411	0.33	%
400	395.14	118.6201	0.14	%
450	445.80	303.5478	0.36	%
500	496.45	274.7696	0.33	%

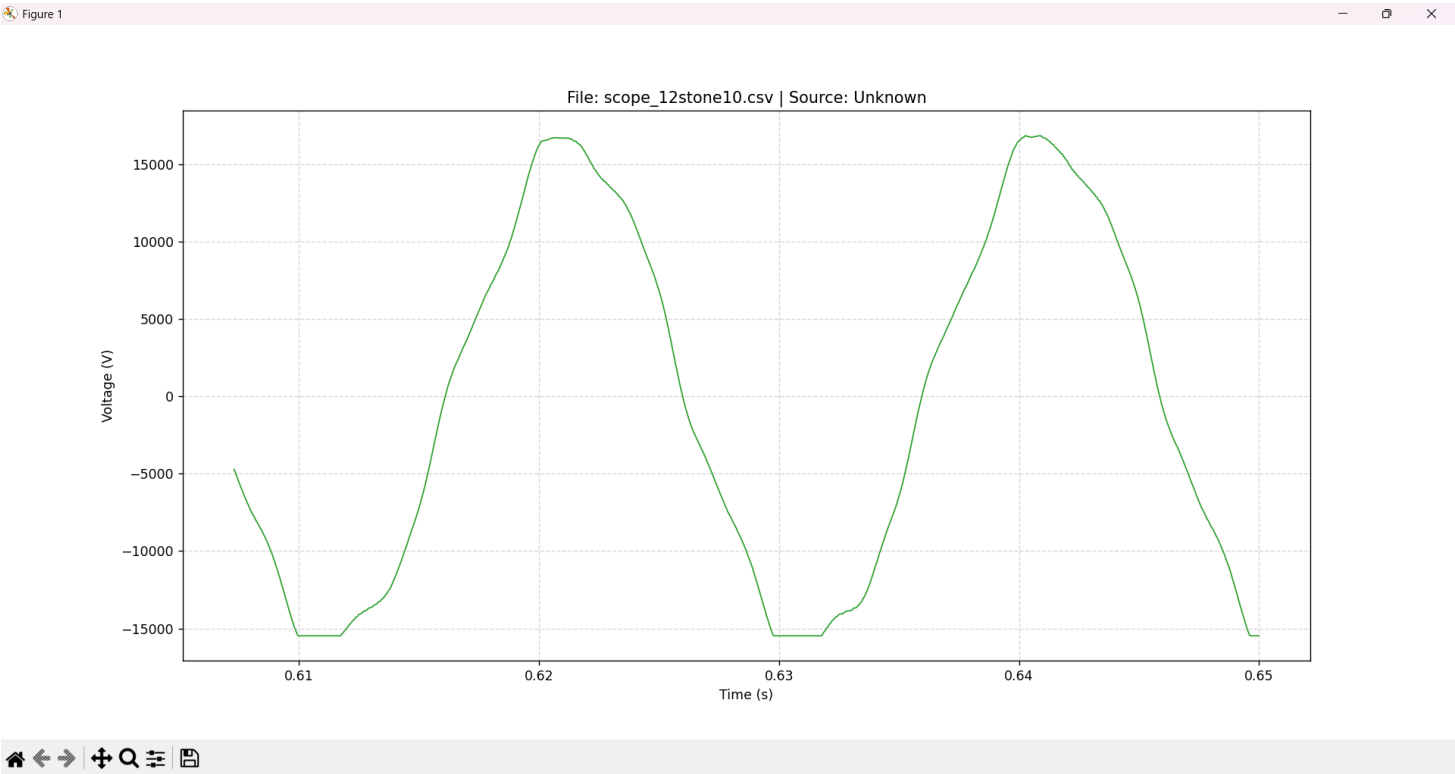
Detected Sampling Rate: 20000.00 Hz

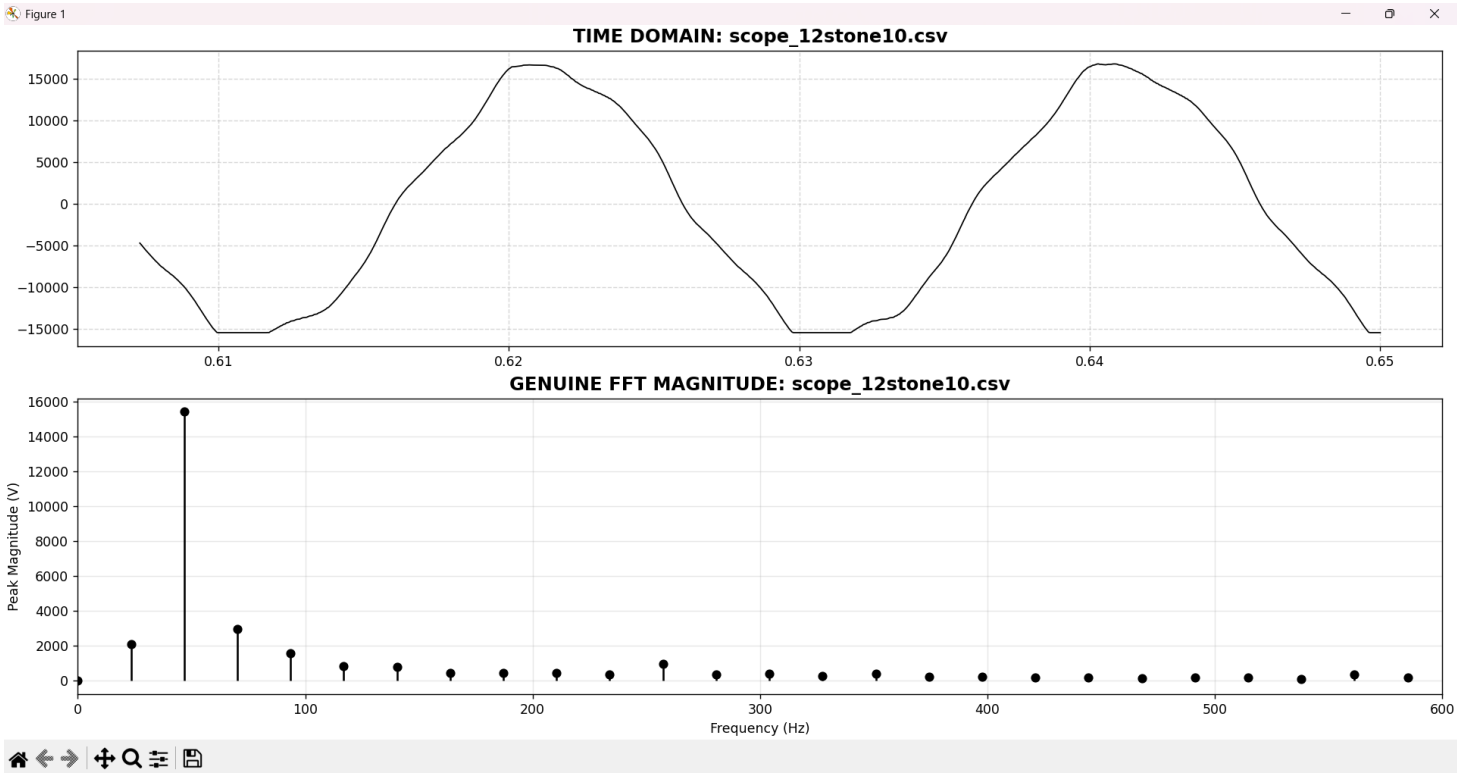


--- EXACT POWER CONTRIBUTION: scope_12stone5.csv ---

Freq (Hz) | Power (V^2/Hz) | Contribution (%)

50.7	1.13e+07	72.73	%
101.3	1.01e+04	0.07	%
152.0	8.15e+03	0.05	%
202.6	1.32e+03	0.01	%
253.3	1.55e+04	0.10	%
304.0	2.28e+03	0.01	%
354.6	3.79e+03	0.02	%
395.1	6.94e+02	0.00	%
445.8	4.55e+03	0.03	%
496.5	3.73e+03	0.02	%



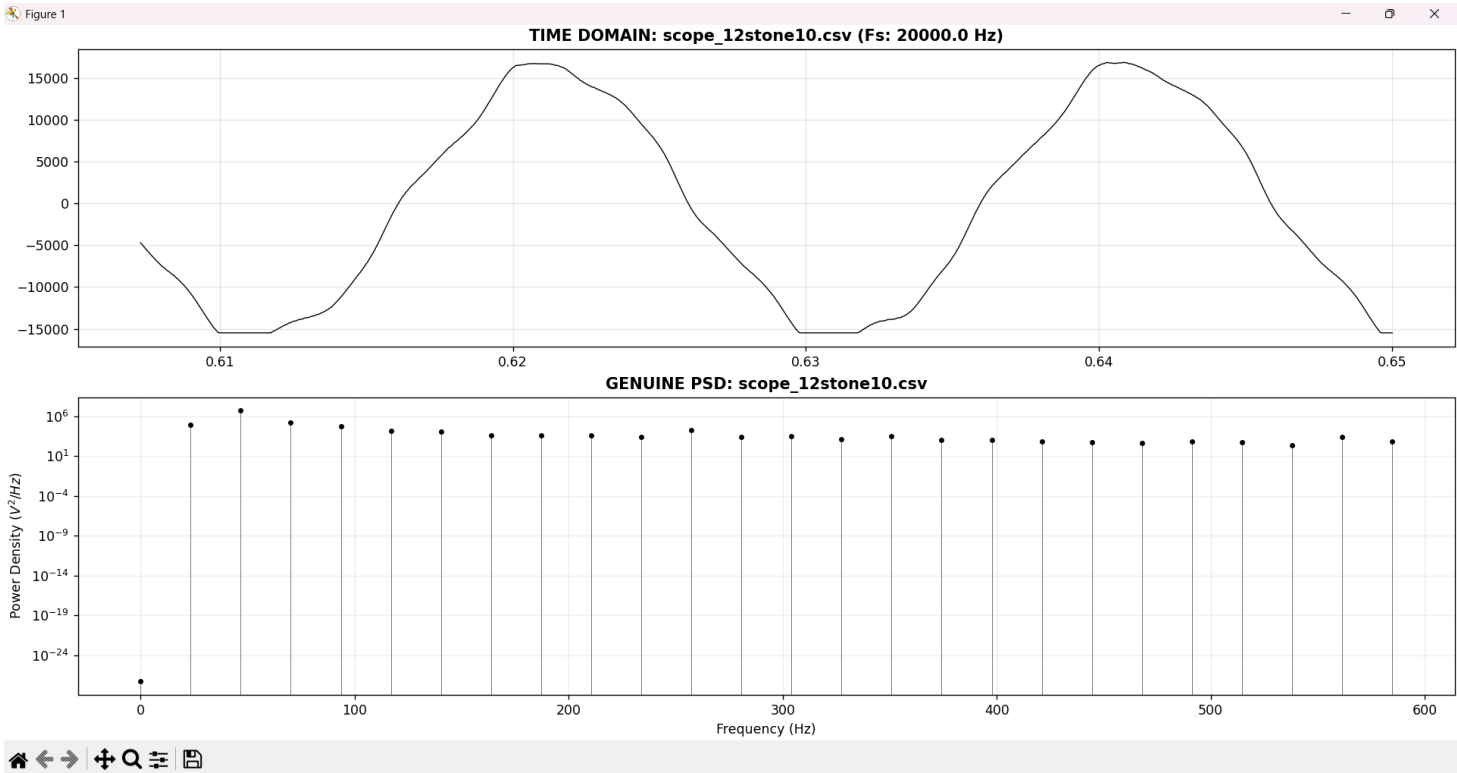


--- GENUINE COORDINATES: scope_12stone10.csv ---

Target (Hz) | Actual (Hz) | Mag (V) | Contrib (%)

50	46.78	15431.2273	38.26	%
100	93.57	1586.1438	3.93	%
150	140.35	781.1054	1.94	%
200	210.53	419.2759	1.04	%
250	257.31	976.4831	2.42	%
300	304.09	370.0649	0.92	%
350	350.88	397.4969	0.99	%
400	397.66	217.9593	0.54	%
450	444.44	155.5015	0.39	%
500	491.23	179.8549	0.45	%

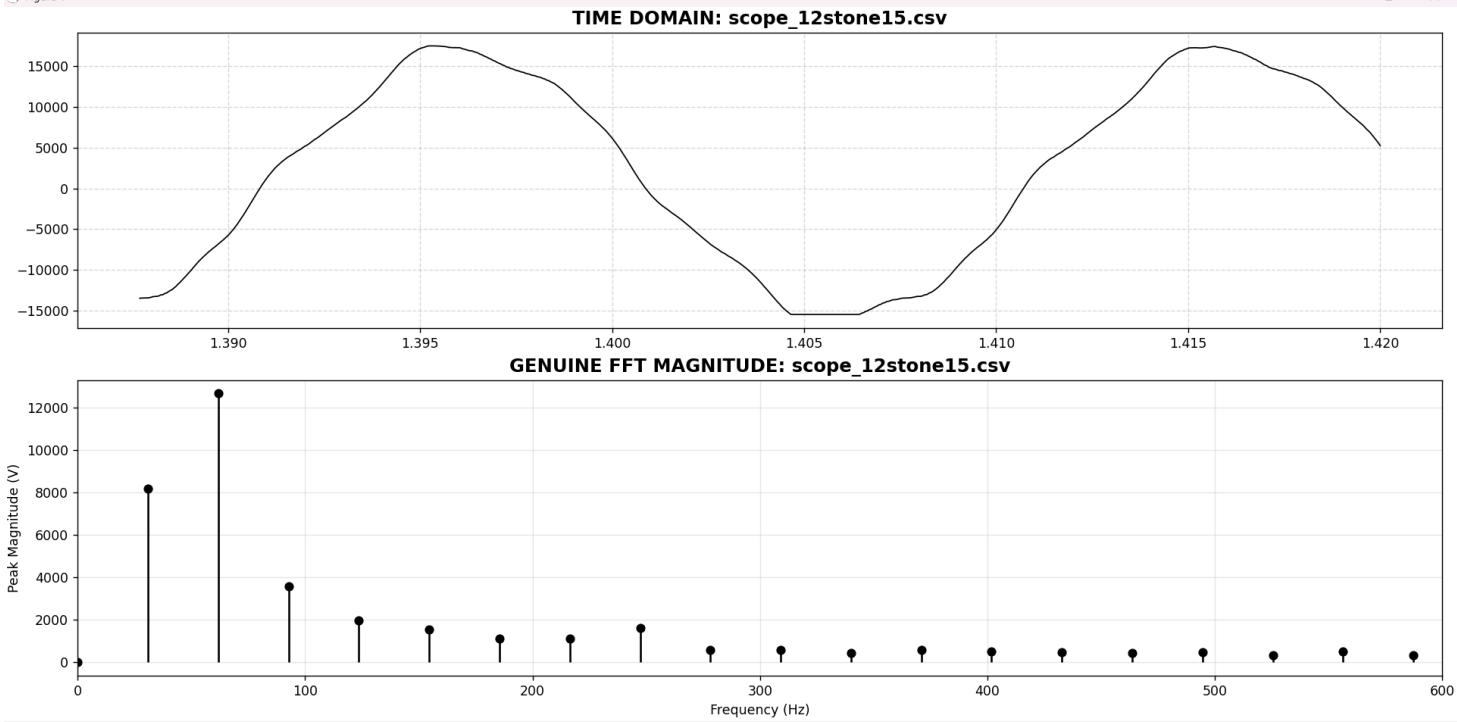
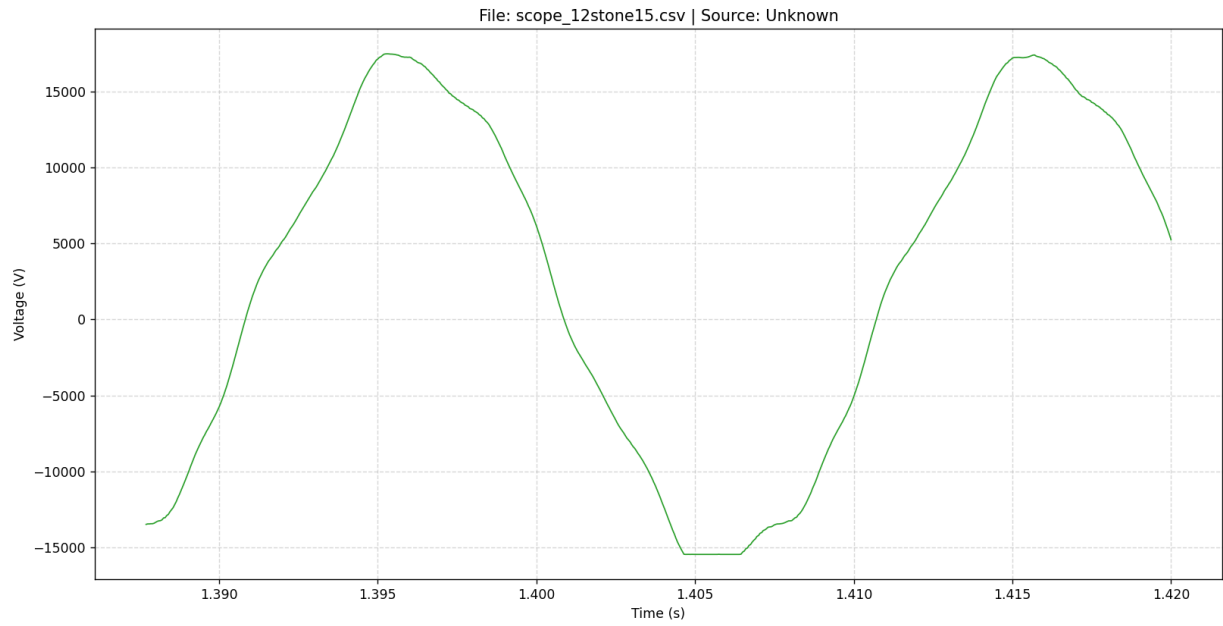
Detected Sampling Rate: 20000.00 Hz



--- EXACT POWER CONTRIBUTION: scope_12stone10.csv ---

Freq (Hz) | Power (V^2/Hz) | Contribution (%)

46.8	5.09e+06	92.43	%
93.6	5.38e+04	0.98	%
140.4	1.30e+04	0.24	%
210.5	3.76e+03	0.07	%
257.3	2.04e+04	0.37	%
304.1	2.93e+03	0.05	%
350.9	3.38e+03	0.06	%
397.7	1.02e+03	0.02	%
444.4	5.17e+02	0.01	%
491.2	6.91e+02	0.01	%



--- GENUINE COORDINATES: scope_12stone15.csv ---

Target (Hz) | Actual (Hz) | Mag (V) | Contrib (%)

50	61.82	12654.9932	23.05	%
100	92.74	3583.8013	6.53	%
150	154.56	1535.9035	2.80	%
200	185.47	1097.4525	2.00	%
250	247.30	1595.1014	2.91	%
300	309.12	576.2099	1.05	%
350	340.03	427.0765	0.78	%
400	401.85	501.5848	0.91	%
450	463.68	422.4445	0.77	%
500	494.59	466.2220	0.85	%

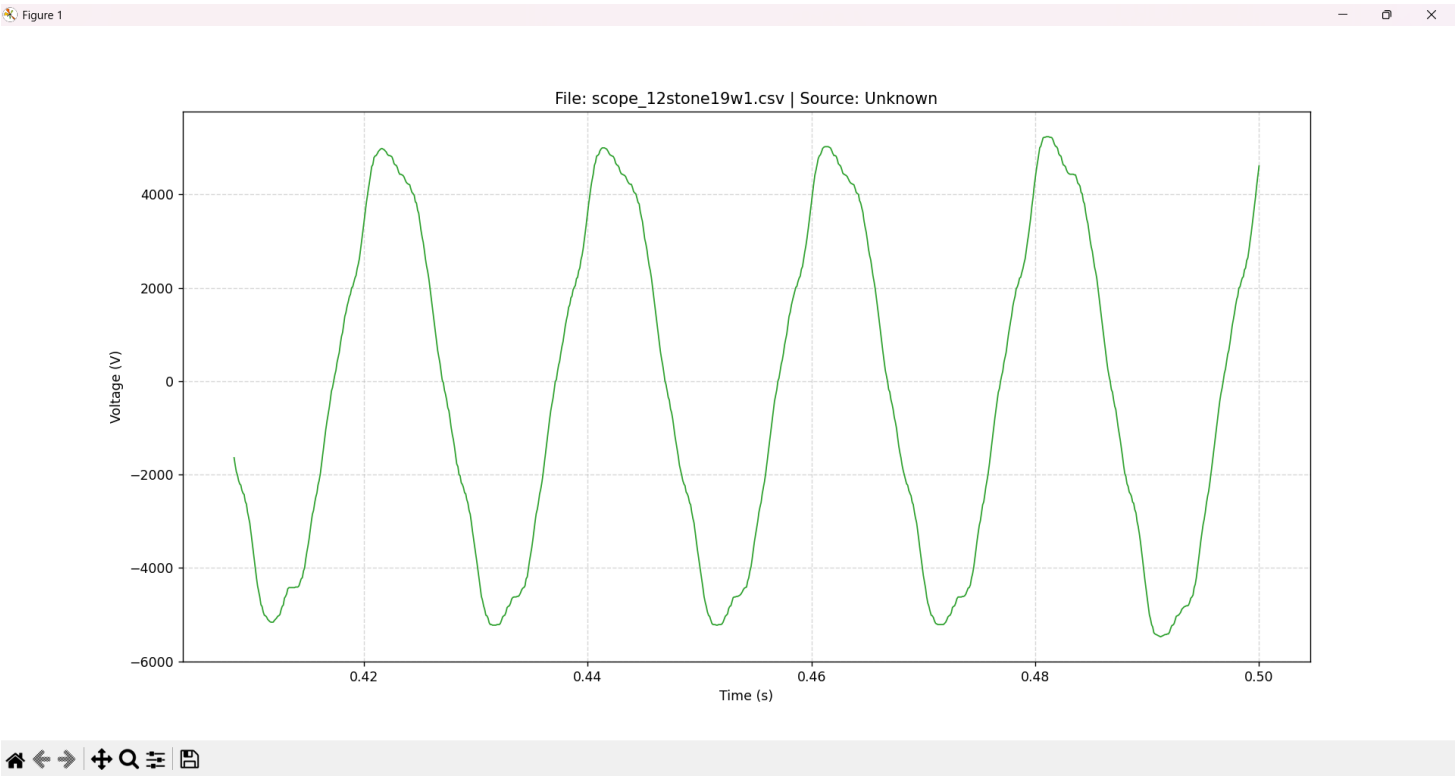
Detected Sampling Rate: 20000.00 Hz

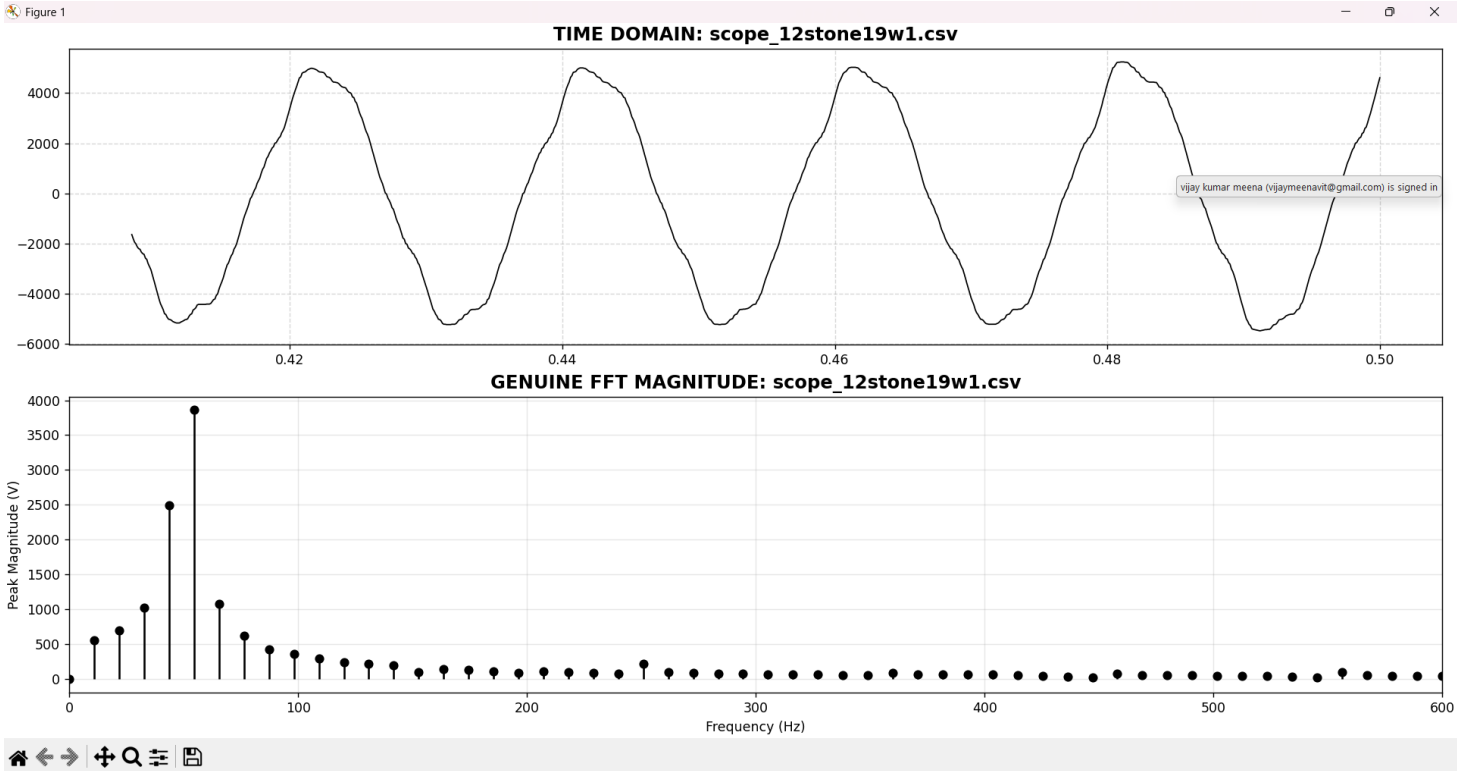


--- EXACT POWER CONTRIBUTION: scope_12stone15.csv ---

Freq (Hz) | Power (V^2/Hz) | Contribution (%)

61.8	2.59e+06	63.20	%
92.7	2.08e+05	5.07	%
154.6	3.82e+04	0.93	%
185.5	1.95e+04	0.48	%
247.3	4.12e+04	1.00	%
309.1	5.37e+03	0.13	%
340.0	2.95e+03	0.07	%
401.9	4.07e+03	0.10	%
463.7	2.89e+03	0.07	%
494.6	3.52e+03	0.09	%





--- GENUINE COORDINATES: scope_12stone19w1.csv ---

Target (Hz) | Actual (Hz) | Mag (V) | Contrib (%)

50	54.53	3860.4933	19.48	%
100	98.15	355.0199	1.79	%
150	152.67	102.6914	0.52	%
200	196.29	92.1576	0.47	%
250	250.82	215.7118	1.09	%
300	305.34	66.6672	0.34	%
350	348.96	53.2719	0.27	%
400	403.49	65.4245	0.33	%
450	447.11	22.9722	0.12	%
500	501.64	47.4284	0.24	%

Detected Sampling Rate: 10000.00 Hz



--- EXACT POWER CONTRIBUTION: scope_12stone19w1.csv ---

Freq (Hz) | Power (V^2/Hz) | Contribution (%)

54.5	6.83e+05	58.92	%
98.1	5.78e+03	0.50	%
152.7	4.84e+02	0.04	%
196.3	3.89e+02	0.03	%
250.8	2.13e+03	0.18	%
305.3	2.04e+02	0.02	%
349.0	1.30e+02	0.01	%
403.5	1.96e+02	0.02	%
447.1	2.42e+01	0.00	%
501.6	1.03e+02	0.01	%

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import os

# Specific file target
file_path = "normal files/scope_12stone19w5.csv"
```



```
target_freqs = [50, 100, 150, 200, 250, 300, 350, 400, 450, 500]
```

```
try:
```

```
# 1. Load and Force Numeric (to fix missing data/NaN issues)
df = pd.read_csv(file_path, skiprows=26, header=None, usecols=[0, 1], engine='python')
df.columns = ['Time', 'Voltage']
df['Time'] = pd.to_numeric(df['Time'], errors='coerce')
df['Voltage'] = pd.to_numeric(df['Voltage'], errors='coerce')
df = df.dropna()
```

```
# 2. Get Exact Timing and FFT
dt = np.median(np.diff(df['Time'].values))
n = len(df)
sig = df['Voltage'].values
sig_centered = sig - np.mean(sig) # DC Removal
```

```
fft_vals = np.fft.rfft(sig_centered)
fft_mag = np.abs(fft_vals) * (2.0 / n)
freqs = np.fft.rfftfreq(n, d=dt)
```

```
# 3. Print the Coordinate Table First
total_mag = np.sum(fft_mag)
print(f"\n--- GENUINE COORDINATES: {os.path.basename(file_path)} ---")
print(f"{'Target (Hz)':<12} | {'Actual (Hz)':<12} | {'Mag (V)':<12} | {'Contrib (%)':<12}")
print(f"{'-' * 65}")
```

```
for t in target_freqs:
    idx = np.abs(freqs - t).argmin()
    actual_f = freqs[idx]
    mag = fft_mag[idx]
    contribution = (mag / total_mag) * 100
    print(f"{'t':<12} | {'actual_f':<12.2f} | {'mag':<12.4f} | {'contribution':<12.2f}%")
```

```
# 4. Generate the Plot
fig, (ax1, ax2) = plt.subplots(2, 1, figsize=(12, 10), layout="constrained")
```

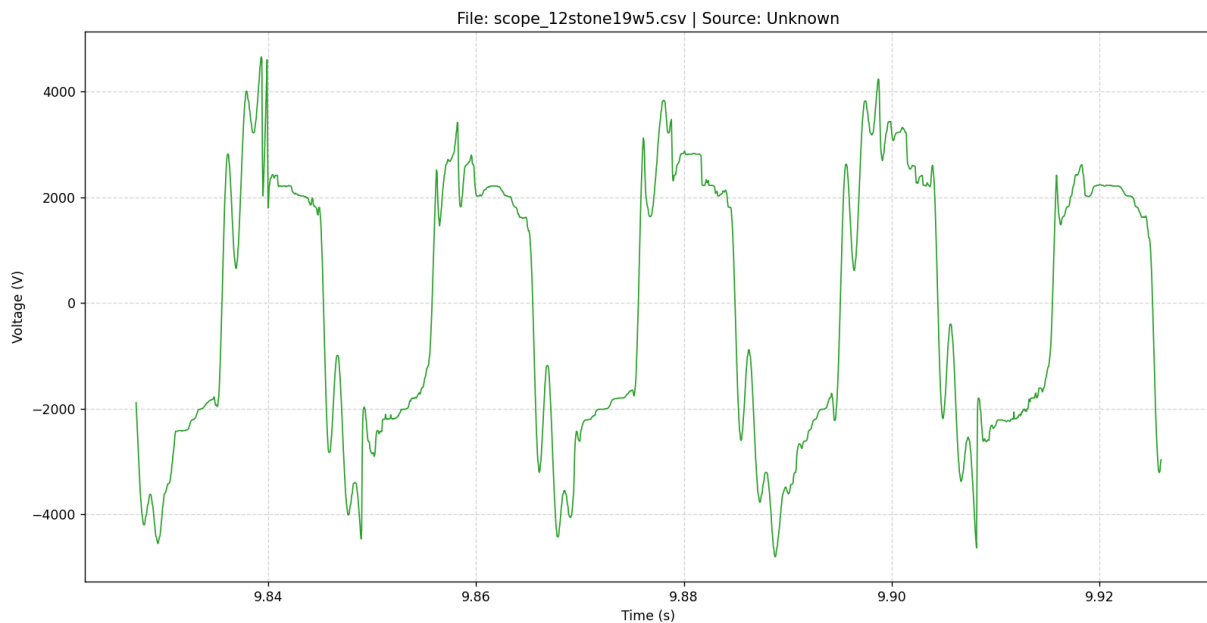
```
# Time Domain
ax1.plot(df['Time'], df['Voltage'], color='black', linewidth=1)
ax1.set_title(f"TIME DOMAIN: {os.path.basename(file_path)}", fontweight='bold')
ax1.grid(True, alpha=0.3)
```

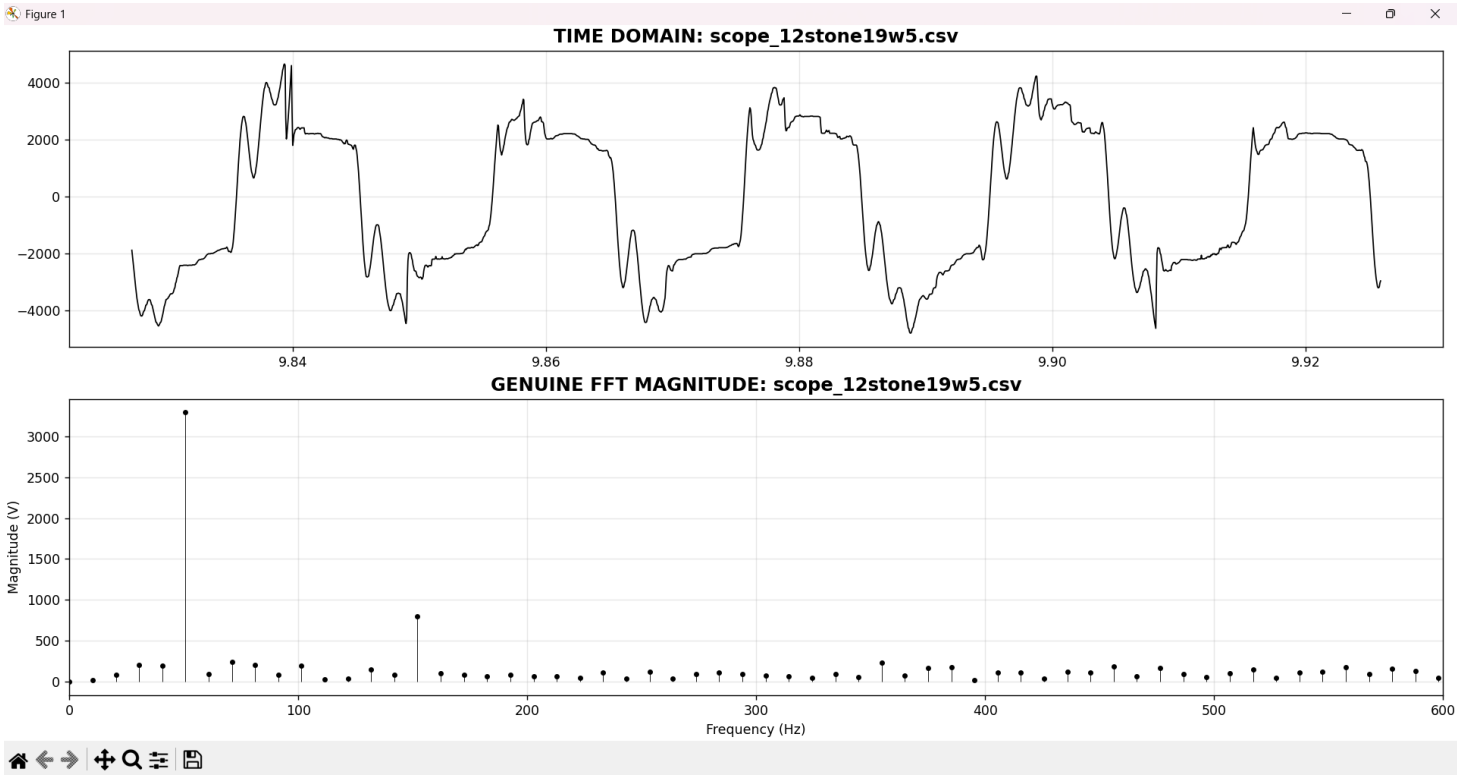
```
# Frequency Domain (Discrete Bins)
markerline, stemlines, baseline = ax2.stem(freqs, fft_mag, basefmt=' ')
plt.setp(markerline, 'markersize', 3, 'color', 'black')
plt.setp(stemlines, 'linewidth', 0.5, 'color', 'black')
ax2.set_xlim(0, 600)
ax2.set_title(f"GENUINE FFT: {os.path.basename(file_path)}", fontweight='bold')
ax2.set_ylabel("Magnitude (V)")
ax2.set_xlabel("Frequency (Hz)")
ax2.grid(True, alpha=0.3)
```

```
# Force the plot to appear
plt.show()
```

```
except Exception as e:
    print(f"Error: {e}")
```

Figure 1

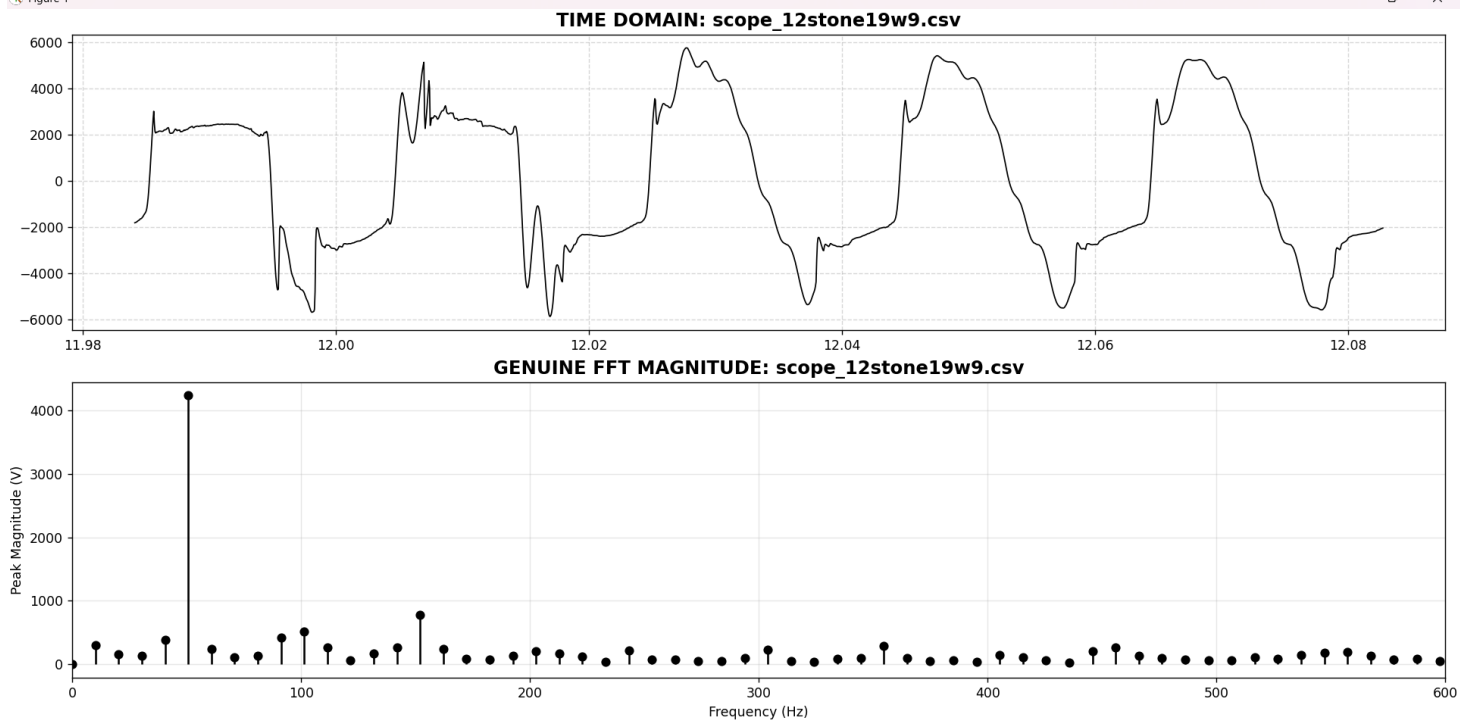
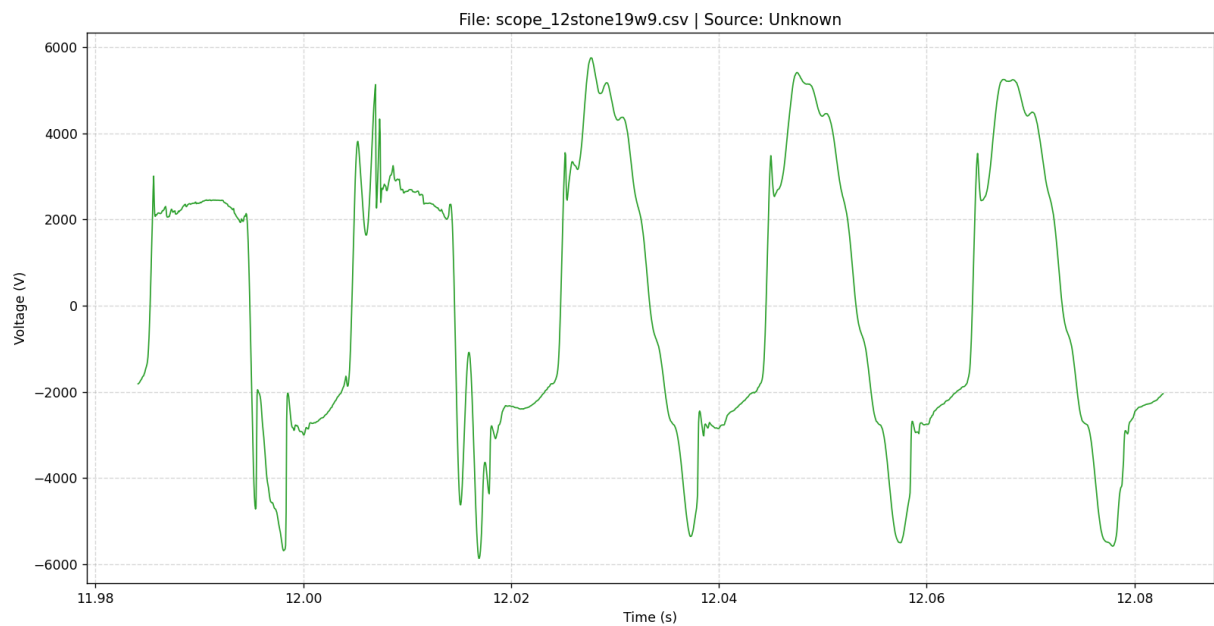




--- GENUINE COORDINATES: scope_12stone19w5.csv ---

Target (Hz) | Actual (Hz) | Mag (V) | Contrib (%)

50	50.68	3292.2279	20.47	%
100	101.37	198.4042	1.23	%
150	152.05	793.9028	4.94	%
200	202.74	64.2129	0.40	%
250	253.42	115.9716	0.72	%
300	304.11	74.9526	0.47	%
350	354.79	234.1653	1.46	%
400	395.34	14.7962	0.09	%
450	446.02	110.4508	0.69	%
500	496.71	53.3648	0.33	%

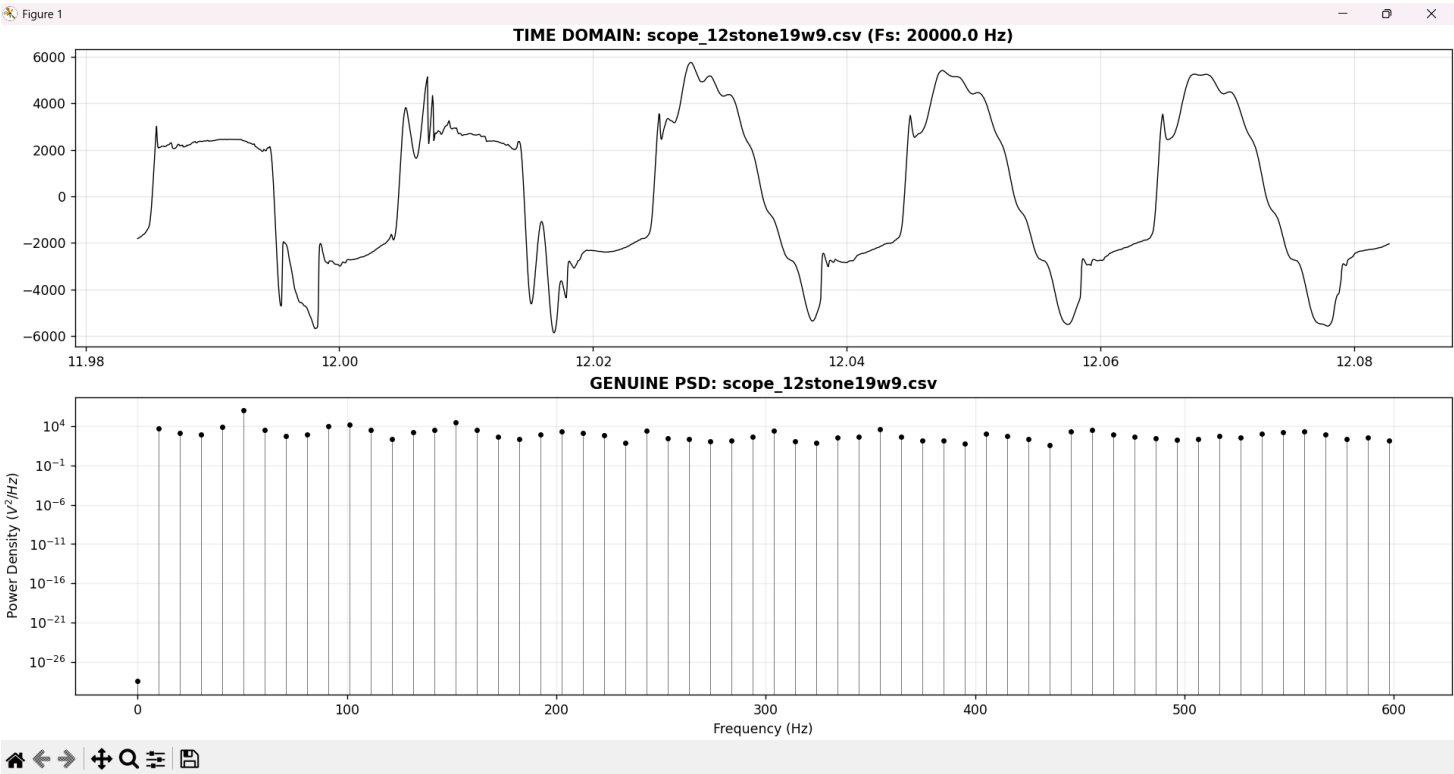


--- GENUINE COORDINATES: scope_12stone19w9.csv ---

Target (Hz) | Actual (Hz) | Mag (V) | Contrib (%)

50	50.66	4235.1478	22.07	%
100	101.32	516.1326	2.69	%
150	151.98	773.7816	4.03	%
200	202.63	207.4769	1.08	%
250	253.29	71.6594	0.37	%
300	303.95	222.0253	1.16	%
350	354.61	286.6125	1.49	%
400	395.14	32.6182	0.17	%
450	445.80	197.2668	1.03	%
500	496.45	61.8725	0.32	%

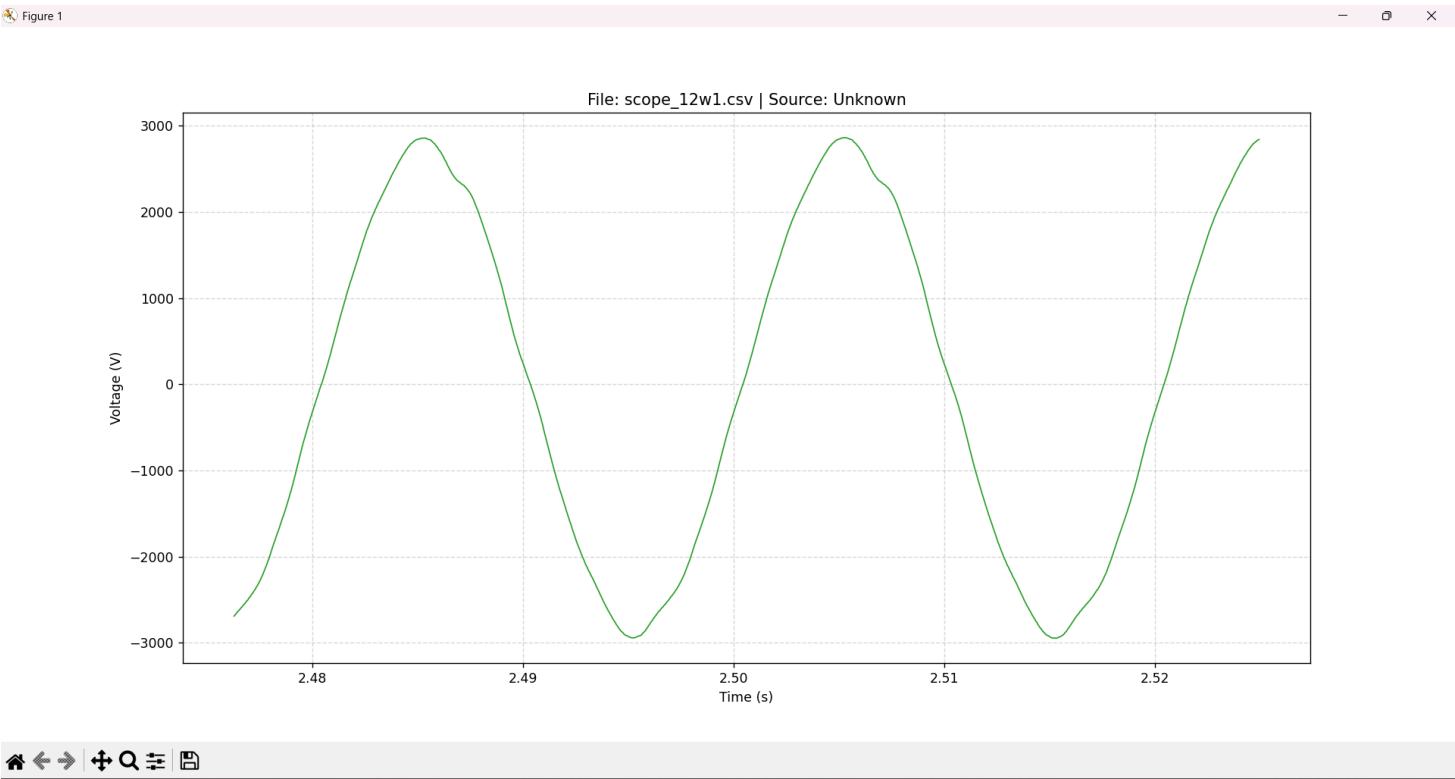
Detected Sampling Rate: 20000.00 Hz

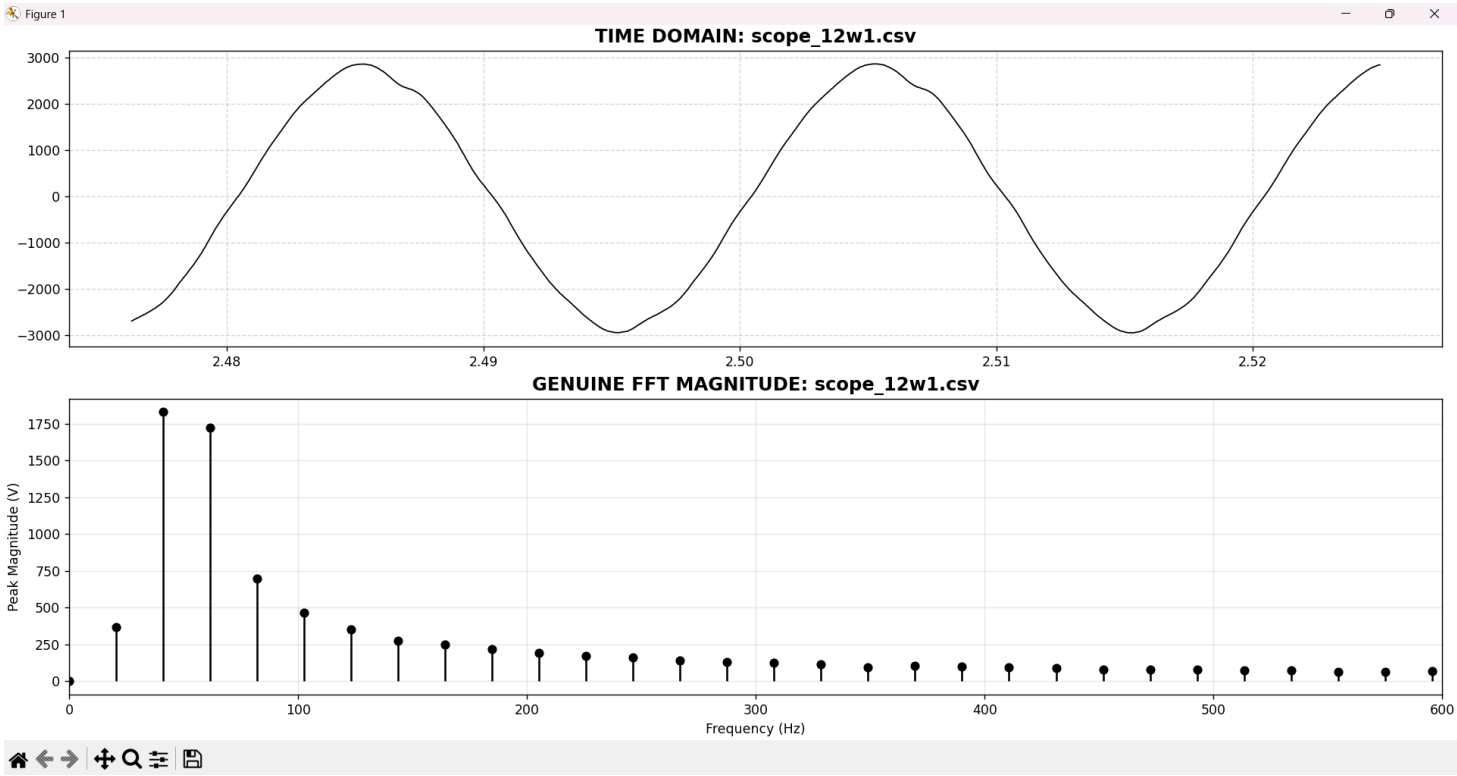


--- EXACT POWER CONTRIBUTION: scope_12stone19w9.csv ---

Freq (Hz) | Power (V^2/Hz) | Contribution (%)

50.7	8.85e+05	88.59	%
101.3	1.31e+04	1.32	%
152.0	2.95e+04	2.96	%
202.6	2.12e+03	0.21	%
253.3	2.53e+02	0.03	%
304.0	2.43e+03	0.24	%
354.6	4.05e+03	0.41	%
395.1	5.25e+01	0.01	%
445.8	1.92e+03	0.19	%
496.5	1.89e+02	0.02	%



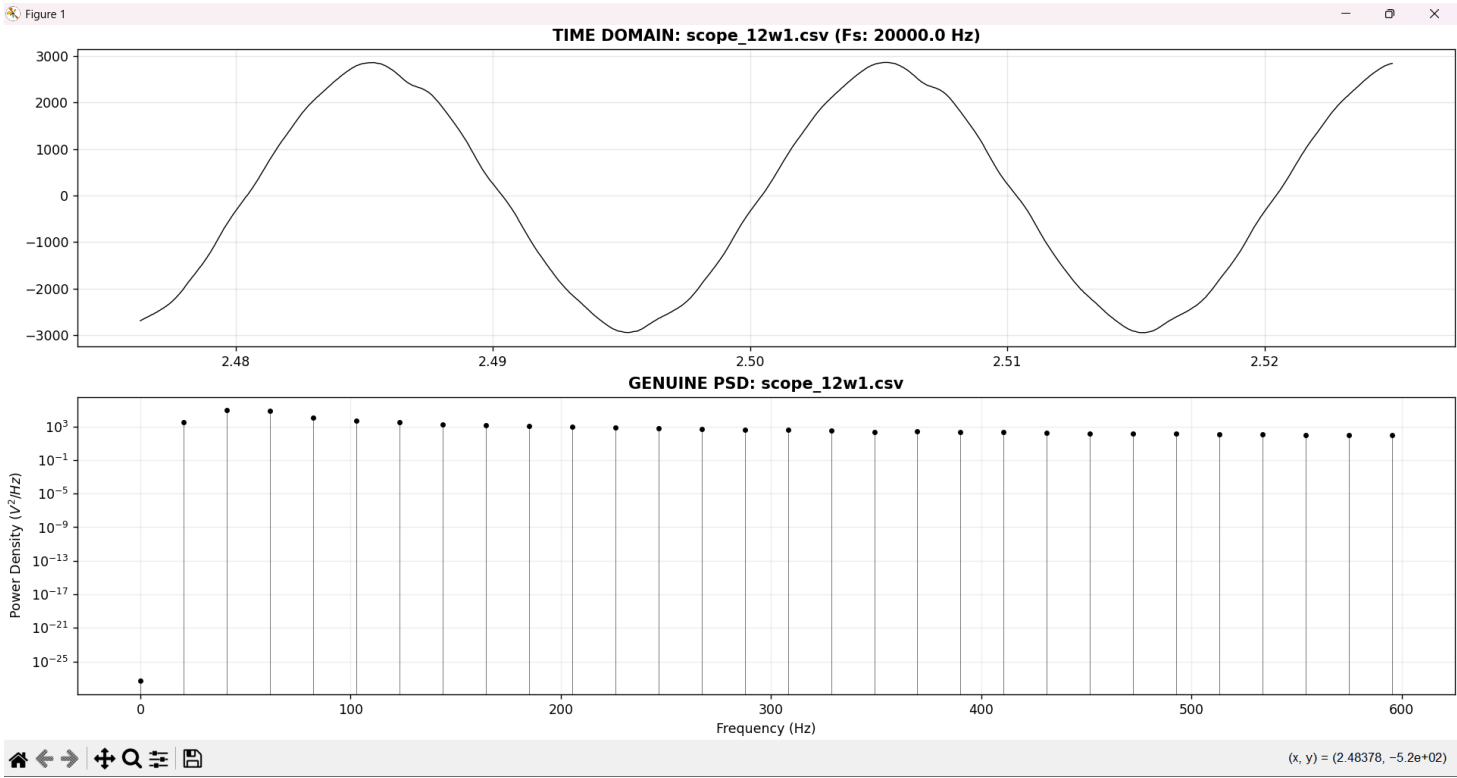


--- GENUINE COORDINATES: scope_12w1.csv ---

Target (Hz) | Actual (Hz) | Mag (V) | Contrib (%)

50	41.07	1829.4539	13.45	%
100	102.67	463.1730	3.41	%
150	143.74	271.6034	2.00	%
200	205.34	191.6354	1.41	%
250	246.41	160.6709	1.18	%
300	308.01	125.3875	0.92	%
350	349.08	91.3915	0.67	%
400	390.14	97.8973	0.72	%
450	451.75	79.4793	0.58	%
500	492.81	78.7430	0.58	%

Detected Sampling Rate: 20000.00 Hz



--- EXACT POWER CONTRIBUTION: scope_12w1.csv ---

Freq (Hz) | Power (V^2/Hz) | Contribution (%)

41.1	8.15e+04	43.54	%
102.7	5.22e+03	2.79	%
143.7	1.80e+03	0.96	%
205.3	8.94e+02	0.48	%
246.4	6.29e+02	0.34	%
308.0	3.83e+02	0.20	%
349.1	2.03e+02	0.11	%
390.1	2.33e+02	0.12	%
451.7	1.54e+02	0.08	%
492.8	1.51e+02	0.08	%